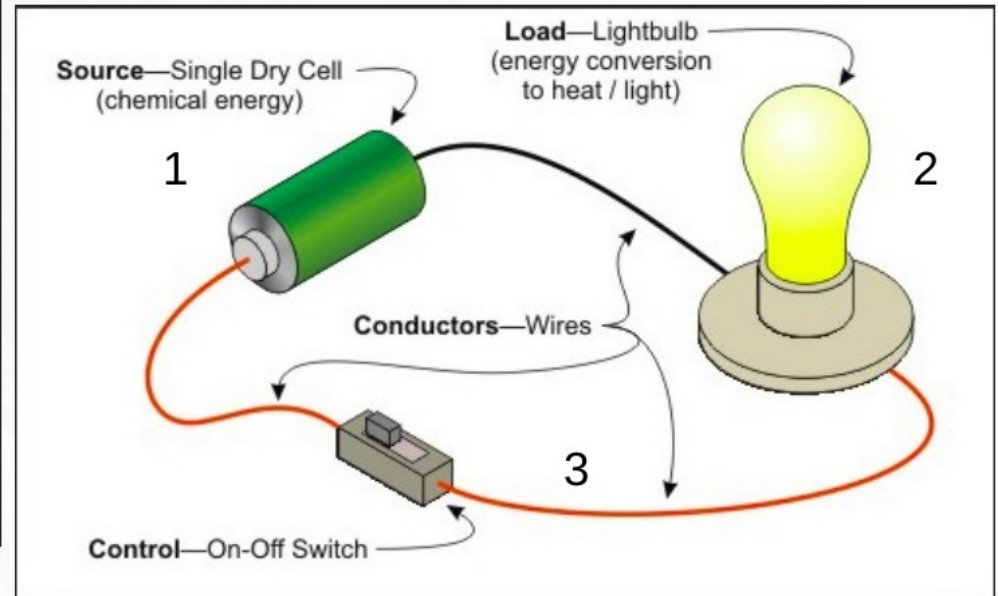


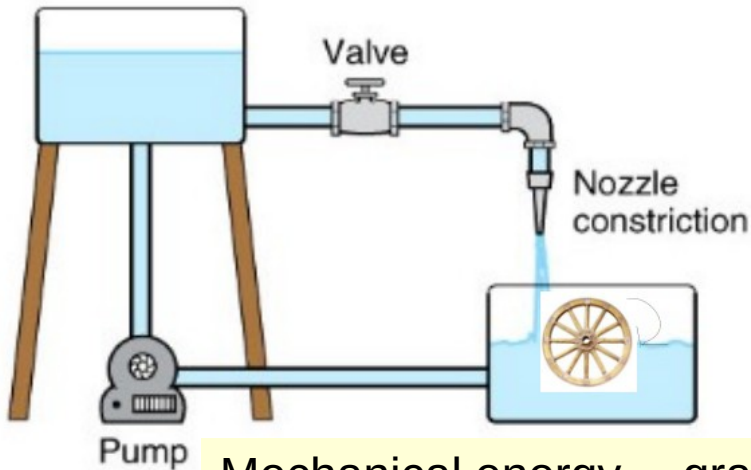
DC circuits - basics



Chemical energy → electrical energy → light

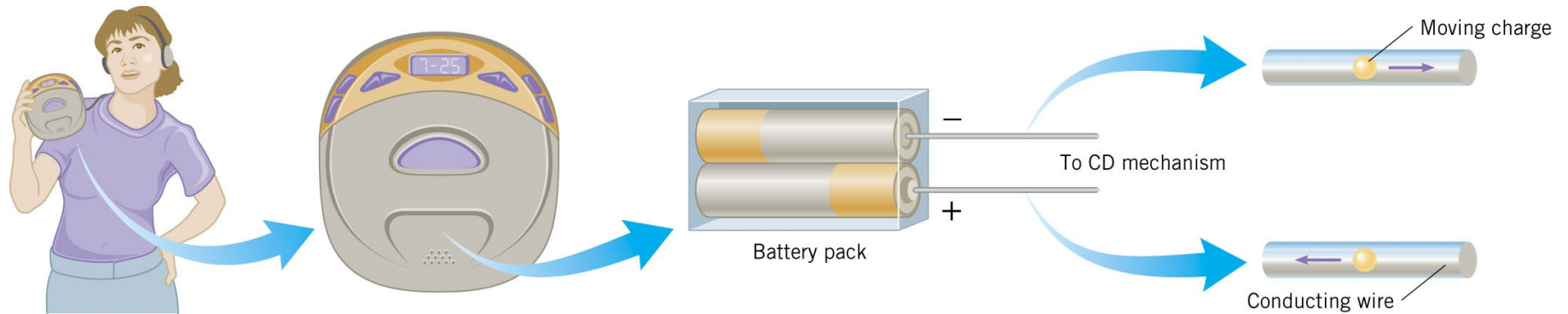
Demo circuit snap kit
Demo circuit

17



Mechanical energy → gravitational energy → KE

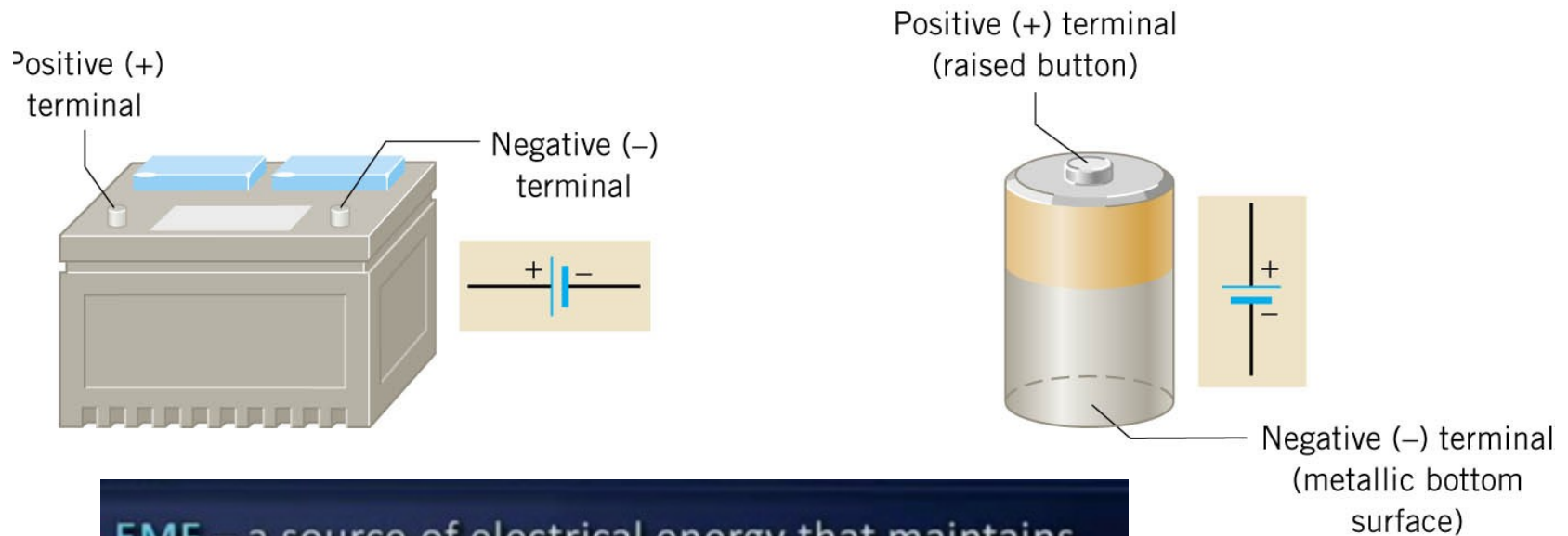
In an electric circuit, an energy source and an energy consuming device are connected by conducting wires through which electric charges move.



Called load

Within a battery, a chemical reaction occurs that transfers electrons from one terminal to another terminal.

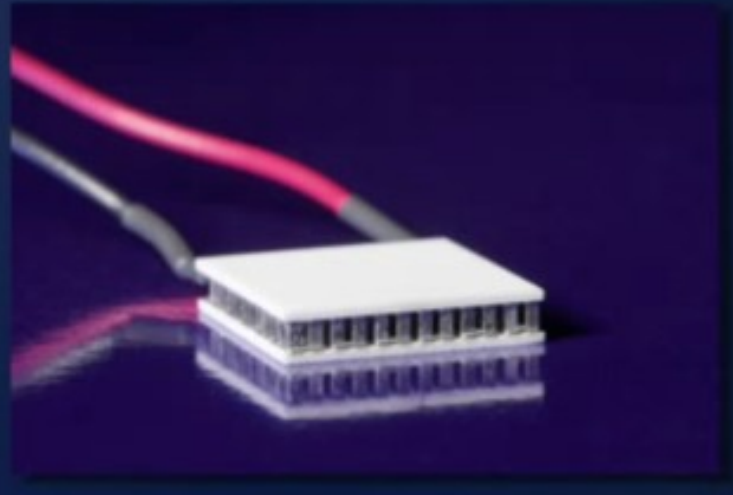
The maximum potential difference across the terminals is called the ***electromotive force (emf)***.



EMF – a source of electrical energy that maintains a constant potential difference across two terminals

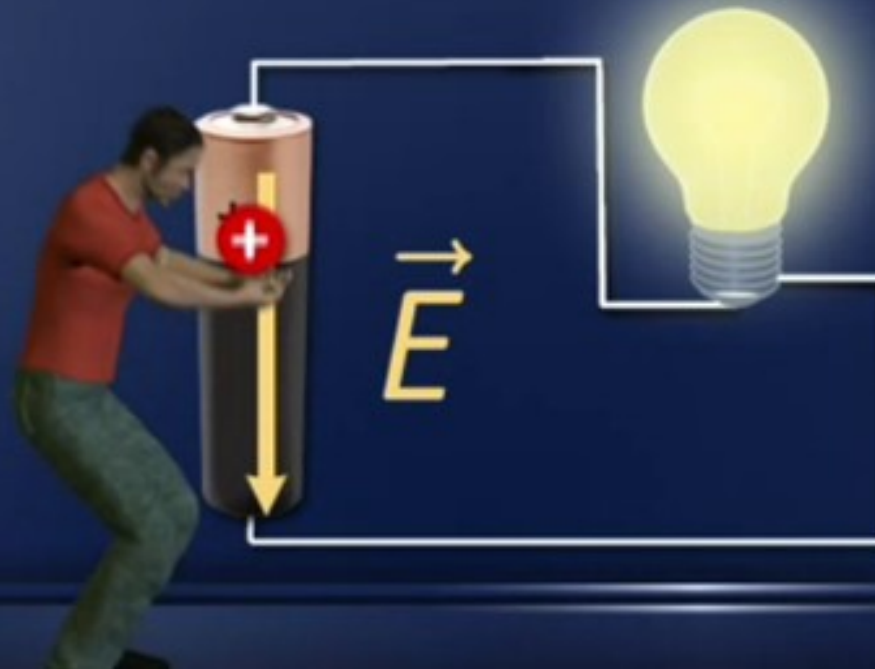
Energizing Circuits

- EMF: a source of electrical energy that maintains a constant potential difference across its two electrical terminals
 - Batteries
 - Generators
 - Power supplies
 - Photovoltaic cells
 - Fuel cells
 - Thermoelectric devices



The Simplest Circuit

- Chemical reactions do work on the charge, “lifting” it against the battery’s internal electric field
 - Work per charge is the battery’s voltage



Electric Current

Electric current is a *flow of charge*

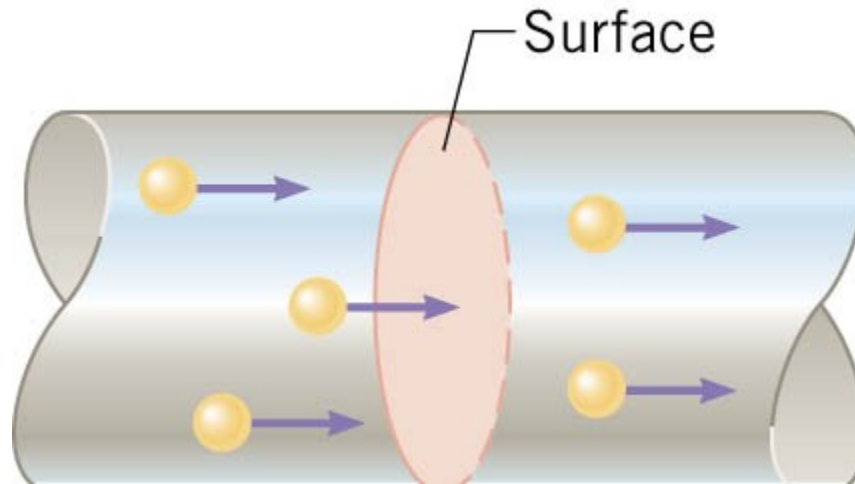
- Quantified as *net* rate of charge crossing an area



Current I to right

- SI unit: coulombs/second
- $1 \text{ C/s} = 1 \text{ ampere (A, amp)}$

The **electric current** is the amount of charge per unit time that passes through a surface that is perpendicular to the motion of the charges.



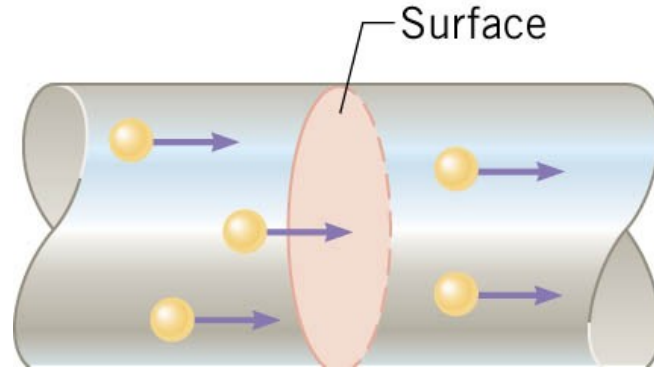
$$I = \frac{\Delta q}{\Delta t}$$

One coulomb per second equals one **ampere** (A).

<https://phet.colorado.edu/en/simulation/circuit-construction-kit-dc>

See how the rate increases when you increase the current

20.1 Electromotive Force and Current



If the charges move around the circuit in the same direction at all times, the current is said to be **direct current (dc)**.

If the charges move first one way and then the opposite way, the current is said to be **alternating current (ac)**.

Charge carriers responding to an electric field :

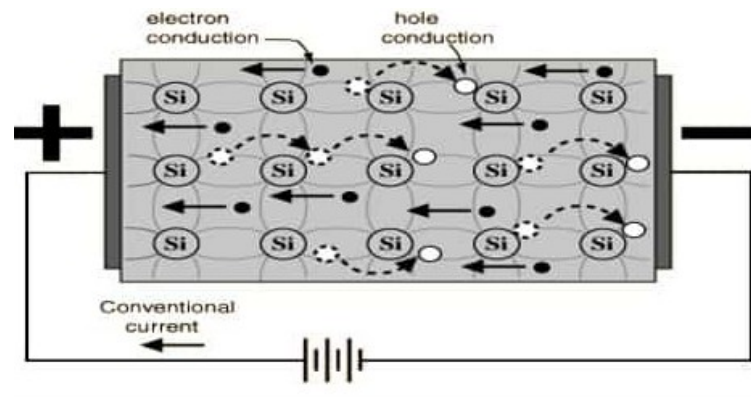
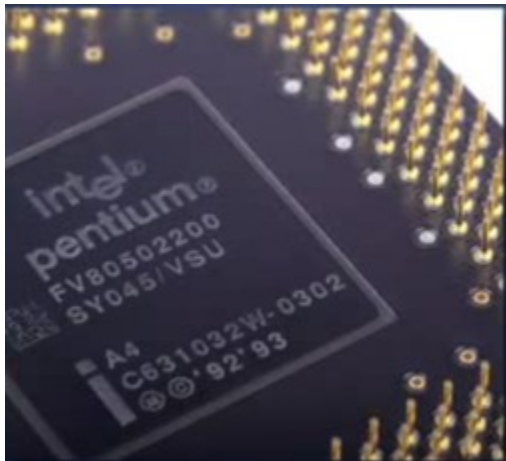
Metal → free electrons

Electrolytes → ions (sport drink)

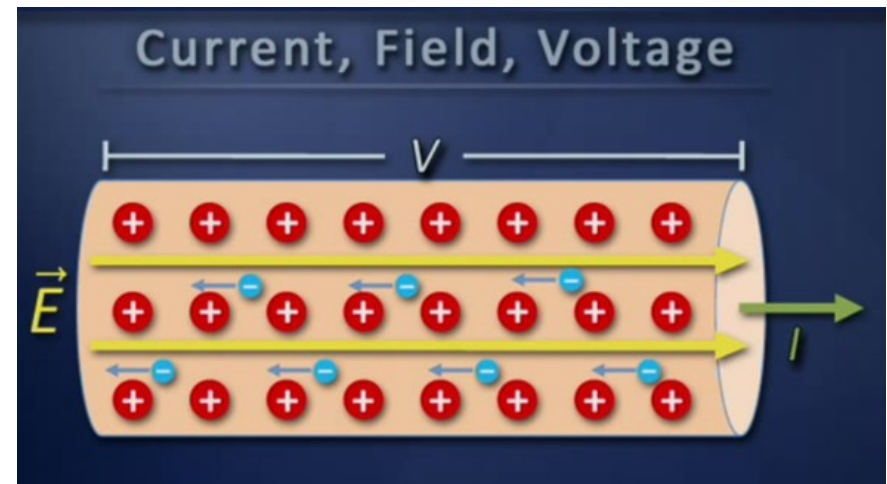
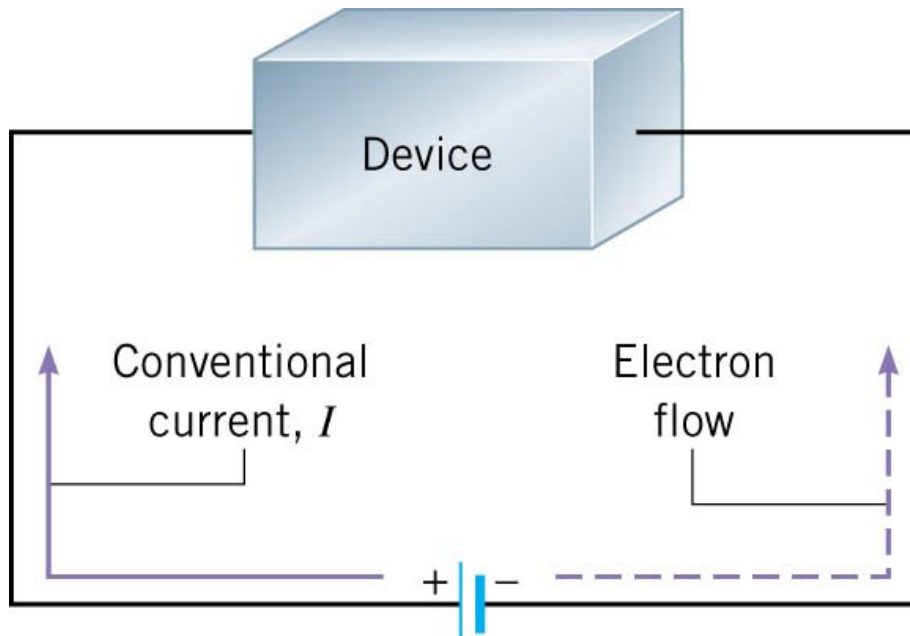
Plasma → ions, electrons (Sun)

Semi conductor → holes or electrons

Supraconductor → electron pairs



Conventional current is the hypothetical flow of positive charges that would have the same effect in the circuit as the movement of negative charges that actually does occur.

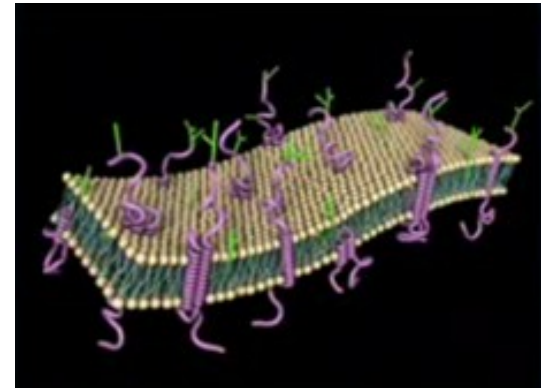




1 amp



1 μ A



1 pA



25 amp



100 amp



1000 amp



2 mA

Physical quantities and equations

t is the time elapsed

Units = seconds (s)

E = total energy consumed/produced.

Units = joules(J)

Q = amount of charge .

Units = coulombs (C)

I = current or amount of charge per unit second = number of charge/ time = Q/t

Units: Amps(A) or equivalent to coulombs/second (C/s)

V or ΔV = difference of potential or voltage or tension= Energy per unit charge = E/Q

Units = volts (V) equivalent to joules/coulomb (J/C)

P = power = energy consumed/produced per unit second = E/t

Units = watts(W) or joules/second(J/s)

P or Power is energy consumed (load) or produced (power supply) every second.

voltage is energy per 1 coulomb. current is number of coulombs per second

So energy/second = $P = \text{voltage} \times \text{current}$ (energy / Coulomb) $\times$ (coulombs / second) cross out coulombs.

R = electrical resistance of a component. Unit is Ohms (Ω)

$$P = V I \quad E = P t \text{ or } P = E/t \quad V = E / Q \quad I = Q/t$$

Voltage across an ohmic component with resistance R **$V = RI$**

I is the current flowing the component. Does not work all the time

Electric Circuits



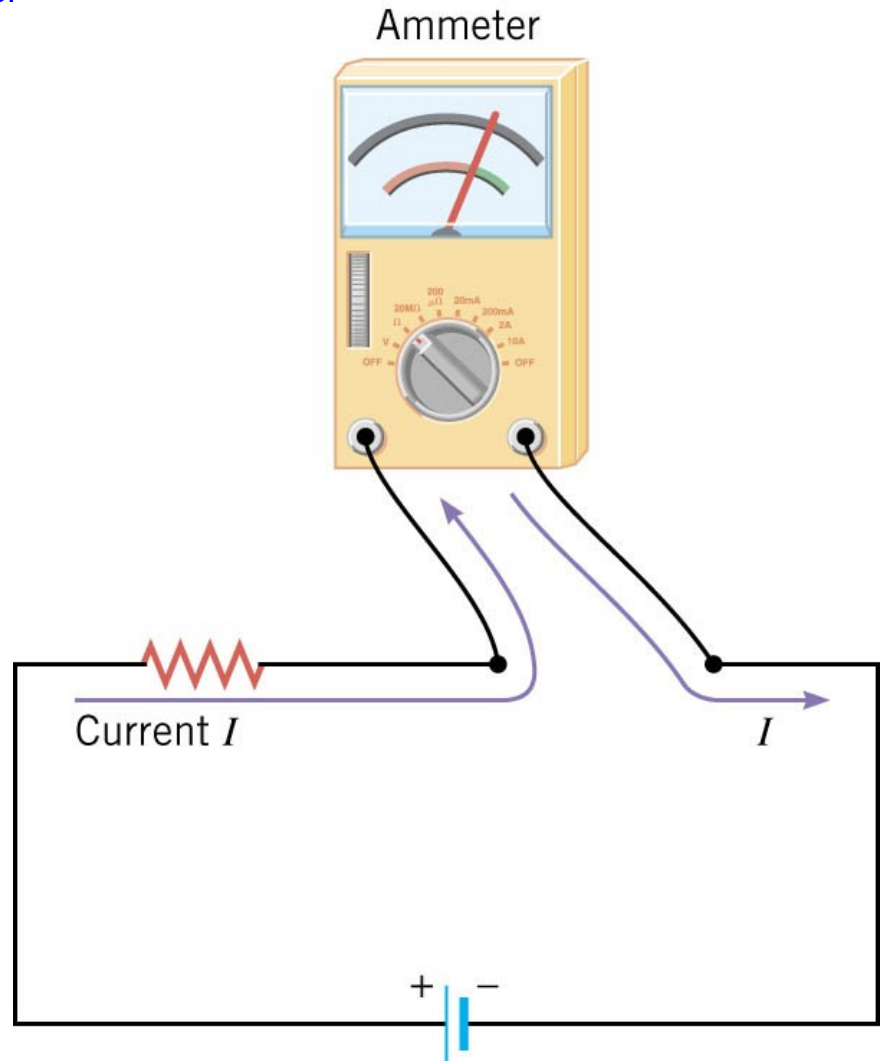
<https://phet.colorado.edu/en/simulations/ohms-law>

<https://phet.colorado.edu/en/simulation/circuit-construction-kit-dc>

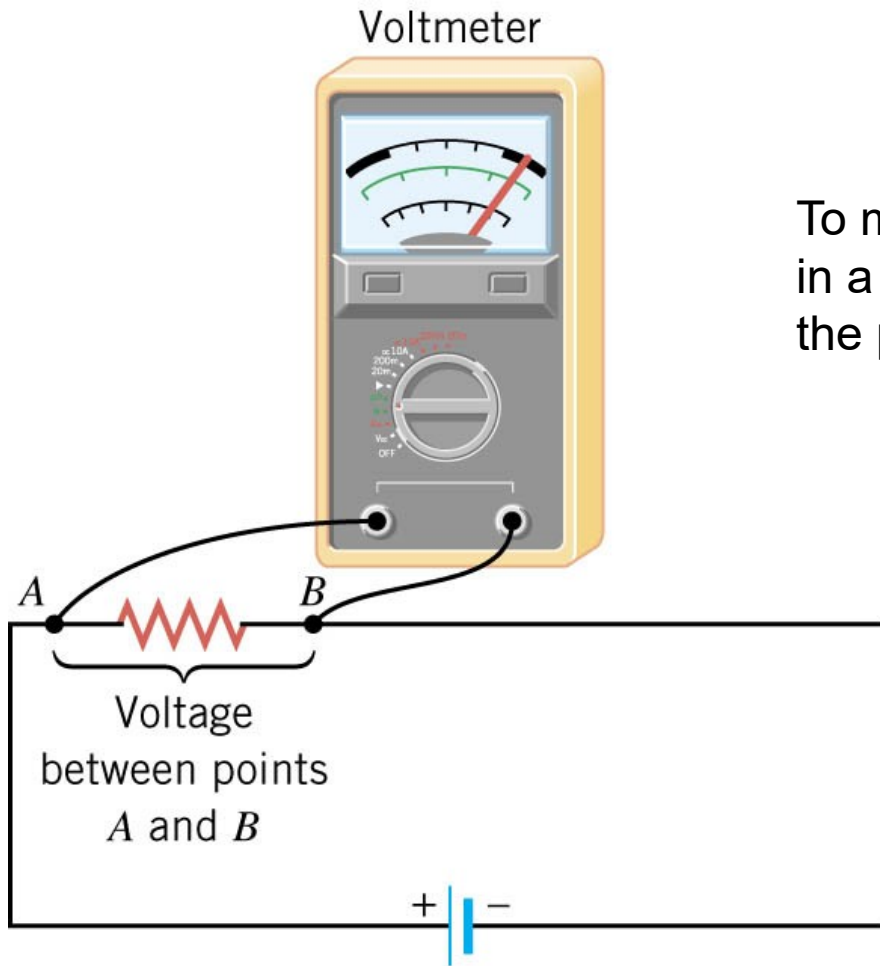
<https://phet.colorado.edu/en/simulation/circuit-construction-kit-ac>

20.11 *The Measurement of Current and Voltage*

An ammeter must be inserted into a circuit so that the current passes directly through it.



Current and Voltage



To measure the voltage between two points in a circuit, a voltmeter is connected between the points.

Example 1 A Pocket Calculator

The current in a 3.0 V battery of a pocket calculator is 0.17 mA. In one hour of operation, (a) how much charge flows in the circuit and (b) how much energy does the battery deliver to the calculator circuit?

Hint: Remember voltage is energy per charge. So energy for 1C of charge Delivered.



Example 1 A Pocket Calculator

The current in a 3.0 V battery of a pocket calculator is 0.17 mA. In one hour of operation, (a) how much charge flows in the circuit and (b) how much energy does the battery deliver to the calculator circuit?

Hint: Remember voltage is energy per charge. So energy for 1C of charge Delivered.

$$(a) \quad \Delta q = I (\Delta t) = (0.17 \times 10^{-3} \text{ A})(3600 \text{ s}) = 0.61 \text{ C}$$

$$(b) \quad \text{Energy} = \text{Charge} \times \frac{\text{Energy}}{\text{Charge}} = (0.61 \text{ C})(3.0 \text{ V}) = 1.8 \text{ J}$$

Electric resistance

Water flows through pipes. Larger the pipe less Resistance to water flowing.

Likewise. Conductors (wires, loads, power supply, Resistors) have resistance. Large resistance: The “ pipe” network get smaller. See app.

The electrical resistance R of an electrical conductor is a measure of the difficulty to pass an electric current through that conductor.

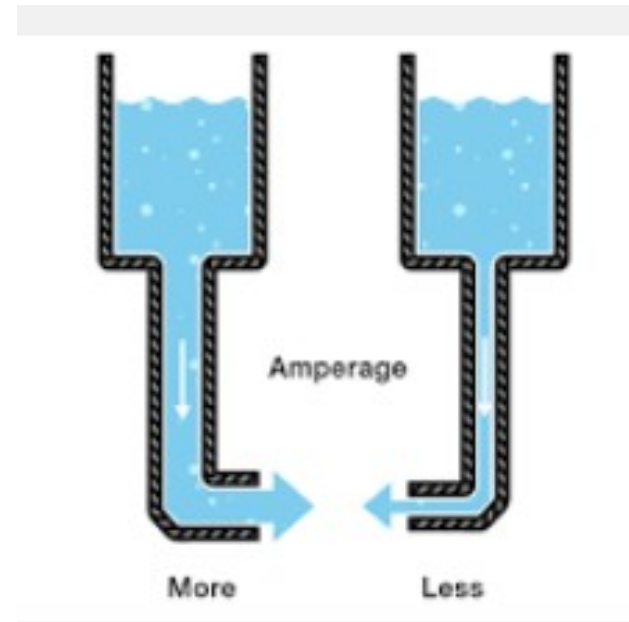
Conductor: wires, loads, resistors

Any component that let the current

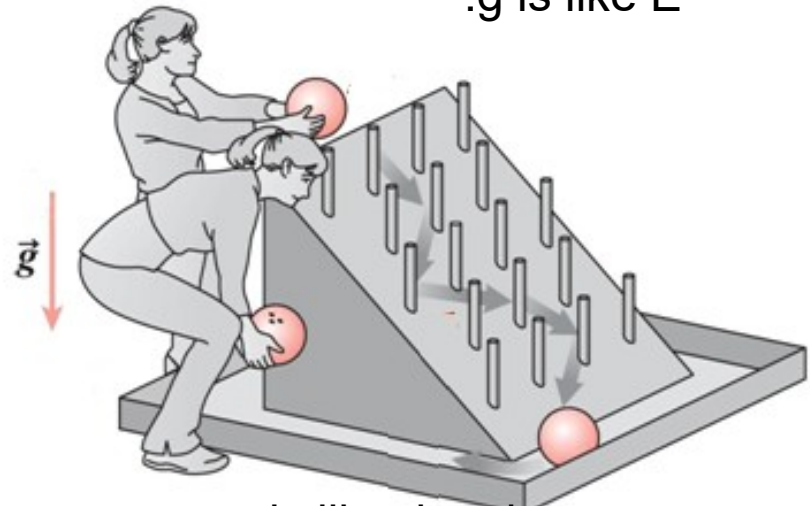
Flow.

Each time you add a load (wire, light bulb, resistor)
, the current decreases.

Each load has a resistance R .
Unit is ohms (Ω)



.g is like E



Mass is like the charge q

$$V = IR.$$

This is Ohm's law.

Voltage across load = resistance x current

Also applies to the whole circuit.

$$\text{Flow} = \text{Pressure} / \text{Resistance}$$

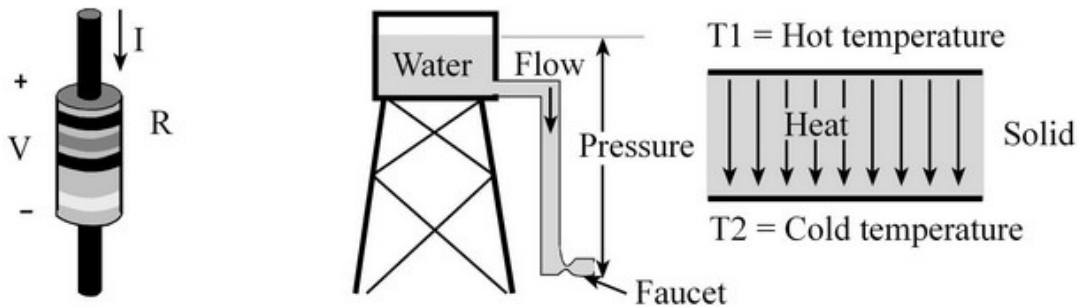
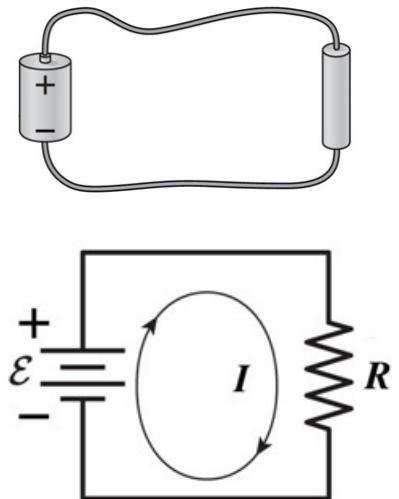


Figure 3.2. Three analogous physical systems demonstrating Ohm's Law.



Physics / cardiology

$$BP = CO \cdot SVR$$

$\downarrow$ Blood Pressure
 $\downarrow$ Cardiac output (blood released per minute)
 $\downarrow$ Systemic vascular Resistance (resistance to blood flow)
 → resistance blood encounters as it flows through vessels

$$CO = HR \cdot SV$$

$\downarrow$ heart rate
 $\downarrow$ stroke volume
 $\frac{\text{beats}}{\text{minute}} \times \frac{\text{Liters}}{\text{beats}}$
 $\rightarrow \frac{\text{Liters}}{\text{minute}} \rightarrow \text{amount of blood sent per minute}$

Blood pressure
 $\uparrow$

L/m
 $\uparrow$

resistance
 $\uparrow$

$$BP = CO \cdot SVR$$

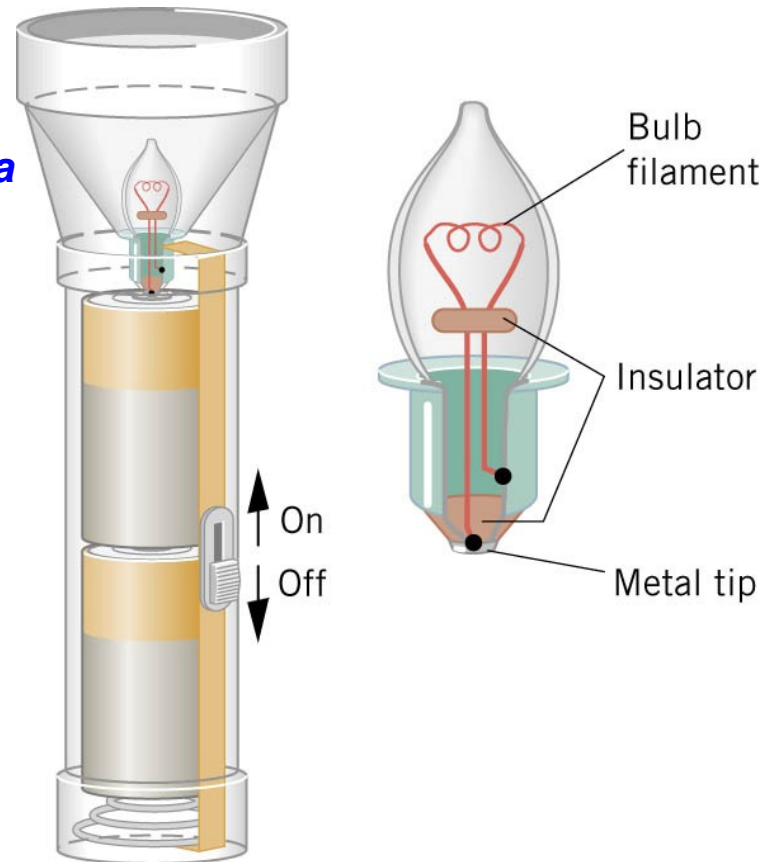
$$V = I \cdot R$$

$\swarrow$ "voltage Pressure"
 $\downarrow$ current (amps) $\frac{C}{s}$
 $\downarrow$ resistance

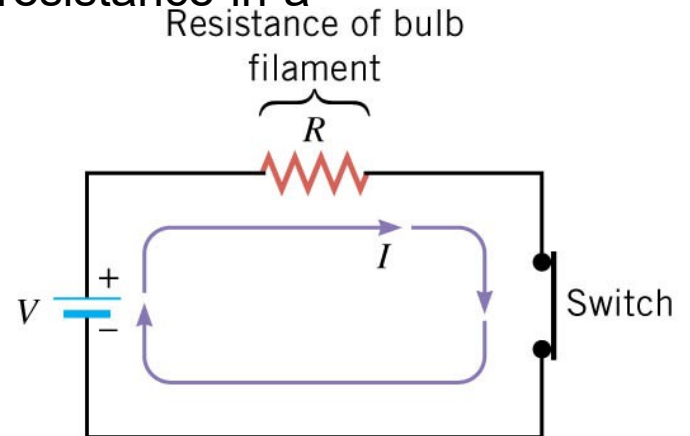
$\hookrightarrow R = \frac{\rho L}{A}$
 $\uparrow$ Blood flow resistance
 $\uparrow \rho$ viscosity (thickness of blood)
 $\uparrow L$ length of blood vessels longer = increased resistance
 $\uparrow A$ Area of the vessels diameter increase Area, decrease resistance

20.2 Ohm's Law

The **resistance** (R) is defined as the ratio of the voltage V applied across a piece of material to the current I through the material.



http://phet.colorado.edu/sims/resistance-in-a-wire/resistance-in-a-wire_en.html



OHM'S LAW

20.2 Ohm's Law

The ratio V/I is a constant, where V is the voltage applied across a piece of material and I is the current through the material:

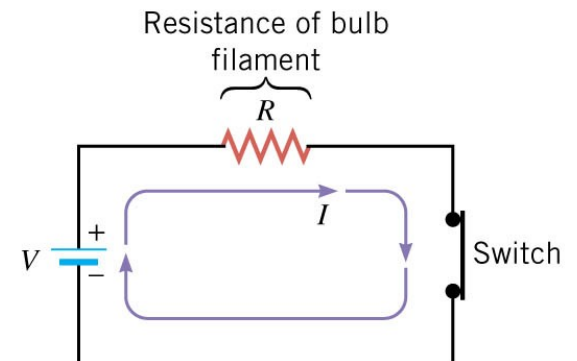
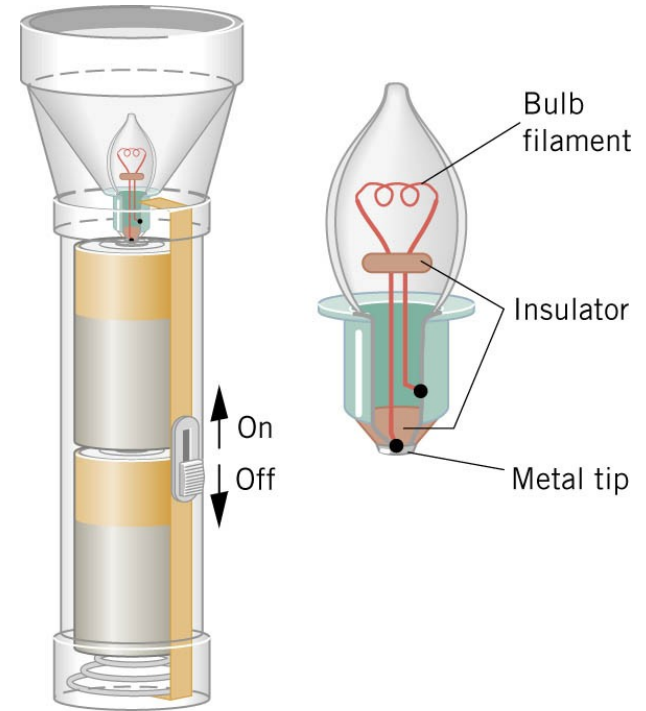
$$\frac{V}{I} = R = \text{constant} \quad \text{or} \quad V = IR$$

Know your terms!

- V is the voltage *across* a resistance R
- I is the current *through* that resistance
- R is the resistance

SI Unit of Resistance: volt/ampere (V/A) = ohm (Ω)

https://phet.colorado.edu/sims/ohms-law/ohms-law_en.html

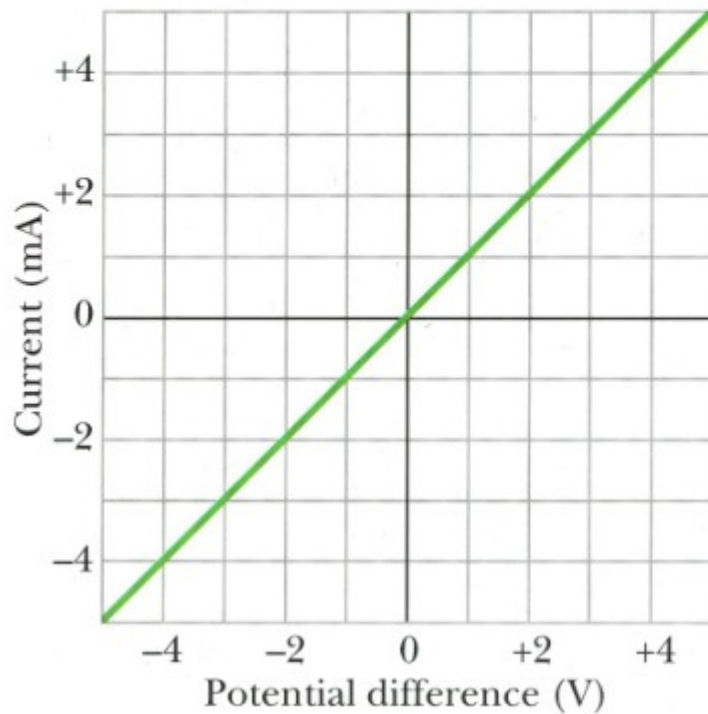


Ohms law is not a nature law.

$$V = RI \text{ with } R = \text{const}$$

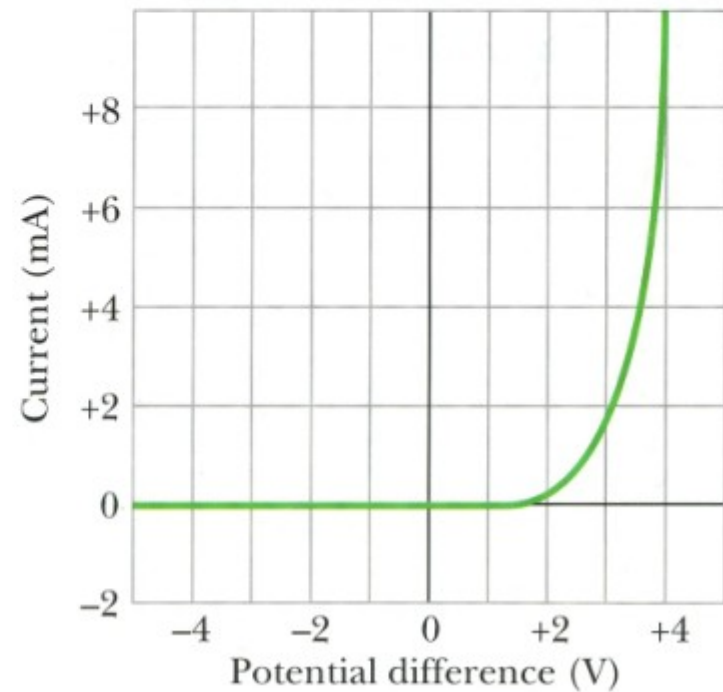
satisfied

RESISTOR 

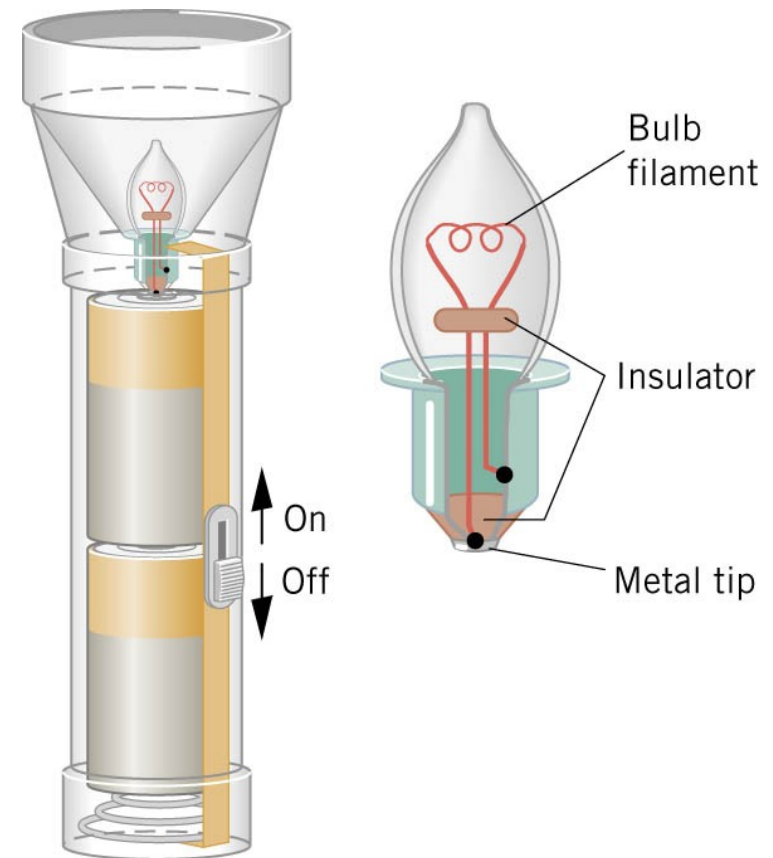
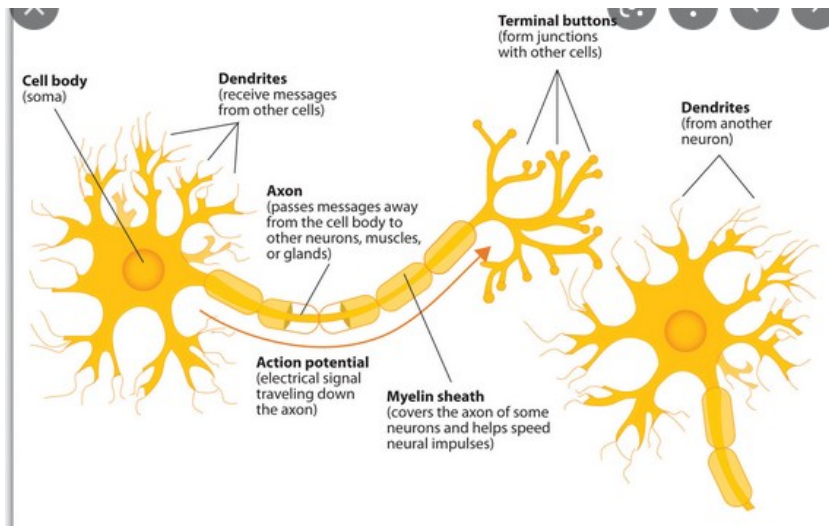


violated

DIODE 

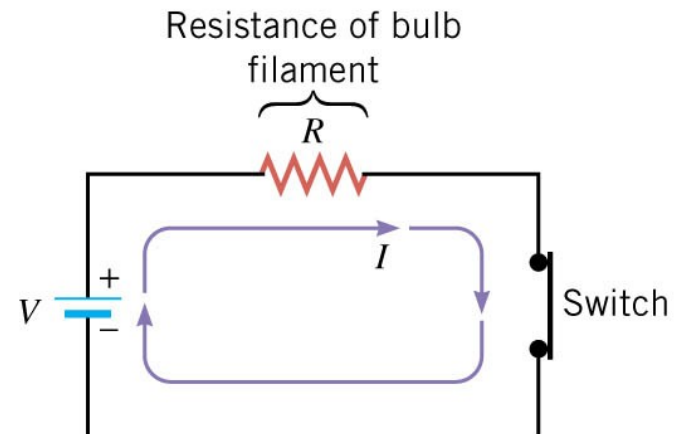


To the extent that a wire or an electrical device offers resistance to electrical flow, it is called a **resistor**.



See book p.818.

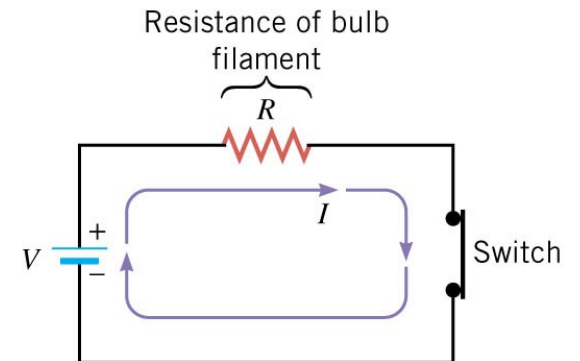
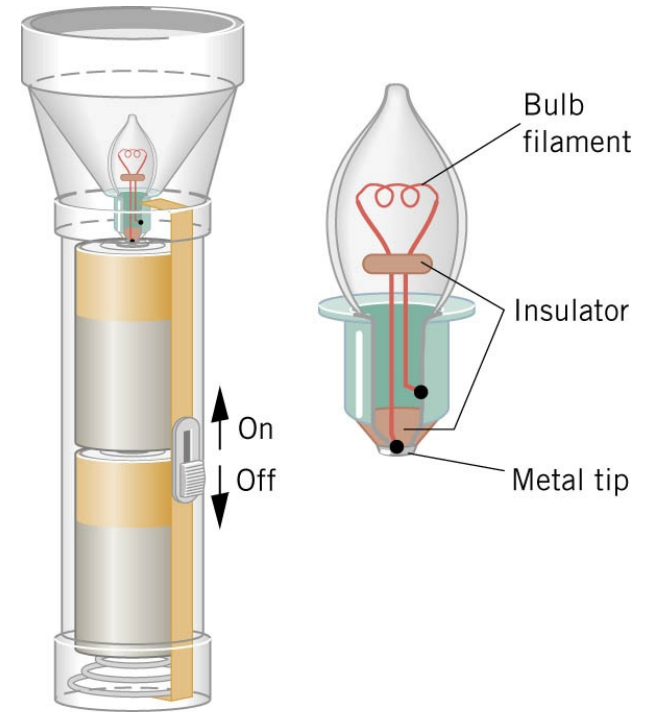
Nerve fiber 1 μm in diameter. The conductive Axon is wrapped in insulating substance called Myelin . The resistance of myelin is high. The electric signal travels along the nerve fiber Rapidly. Because of high resistance of myelin The signal is guided through the fiber.



Example 2 A Flashlight

20.2 Ohm's Law

The filament in a light bulb is a resistor in the form of a thin piece of wire. The wire becomes hot enough to emit light because of the current in it. The flashlight uses two 1.5-V batteries to provide a current of 0.40 A in the filament. Determine the resistance of the glowing filament.

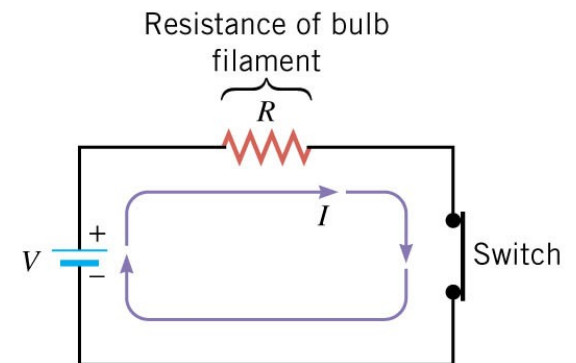
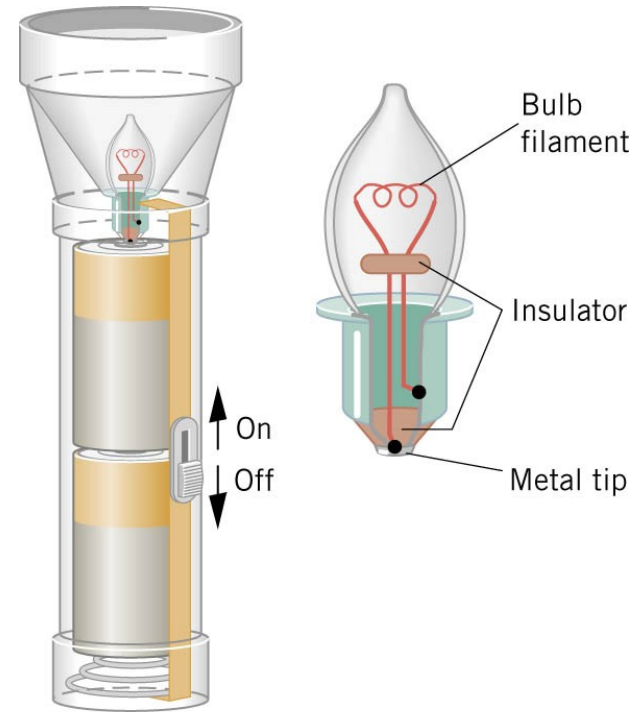


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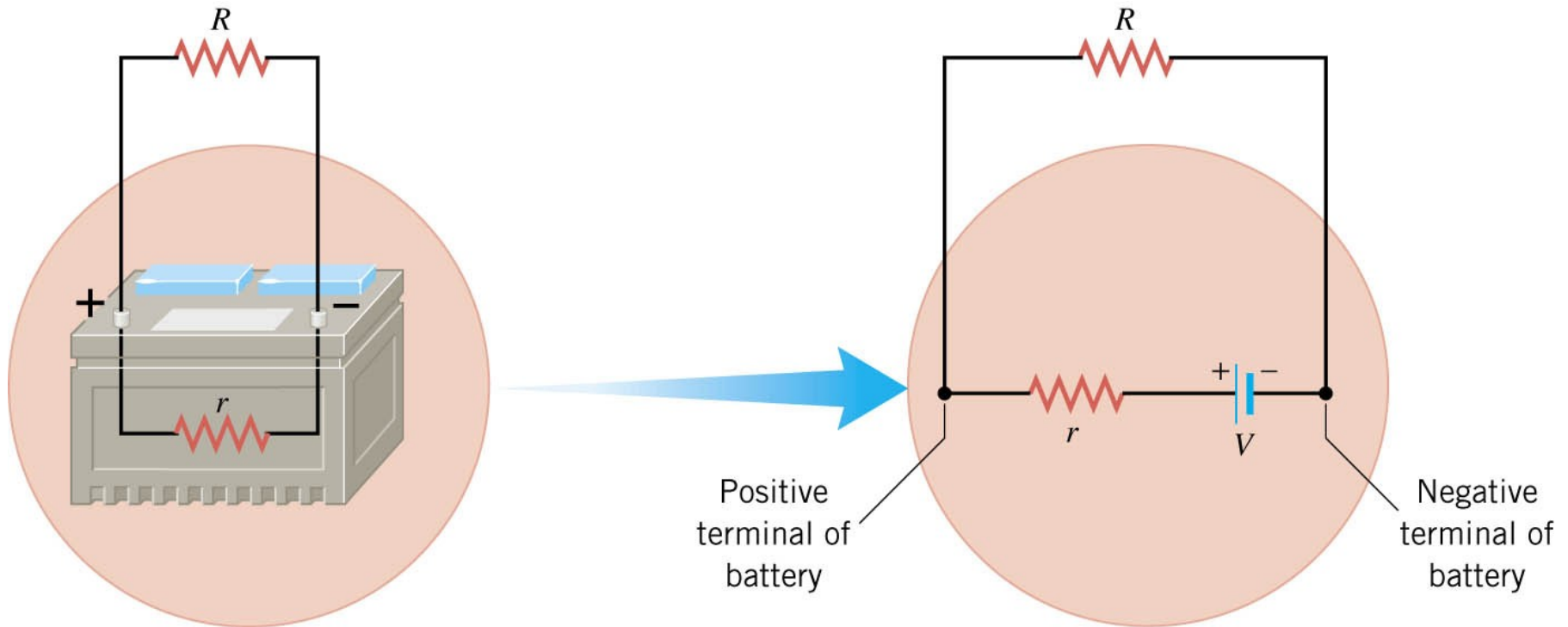
$$R = \frac{V}{I} = \frac{3.0 \text{ V}}{0.40 \text{ A}} = 7.5 \Omega$$



20.9 Internal Resistance

Batteries and generators add some resistance to a circuit. This resistance is called **internal resistance**.

The actual voltage between the terminals of a battery is known as the **terminal voltage**.



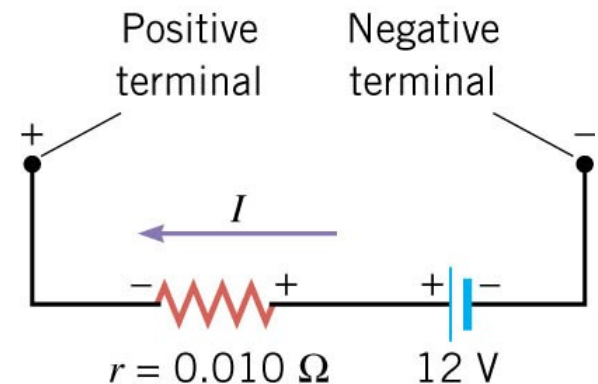
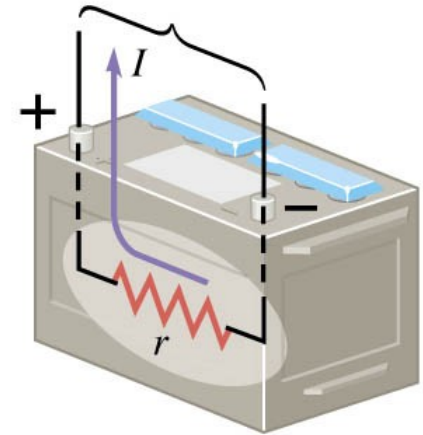
20.9 Internal Resistance

Example 12 The Terminal Voltage of a Battery

The car battery has an emf of 12.0 V and an internal resistance of $0.0100\ \Omega$. What is the terminal voltage when the current drawn from the battery is (a) 10.0 A and (b) 100.0 A? (c) find the rate at which energy is Consumed by the internal resistance.

(a) $V = Ir = (10.0\ \text{A})(0.010\ \Omega) = 0.10\ \text{V}$

To car's electrical system
(ignition, lights, radio, etc.)



20.9 Internal Resistance

Example 12 The Terminal Voltage of a Battery

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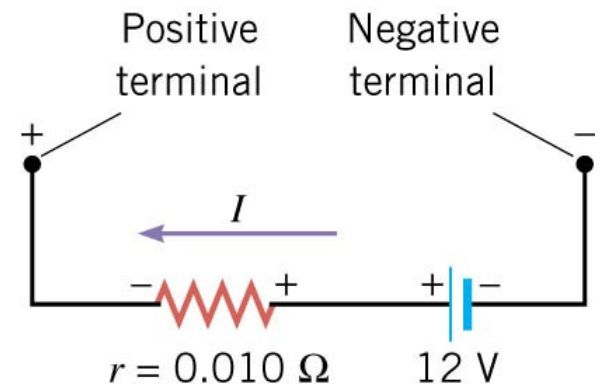
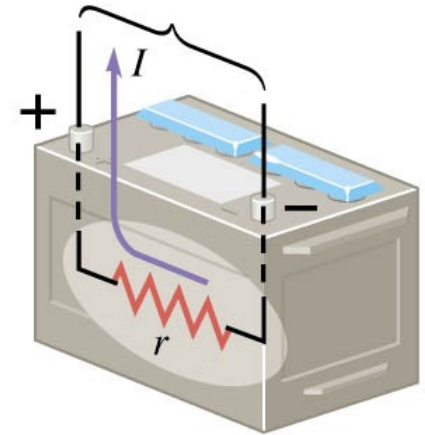
(a) $V = Ir = (10.0\ \text{A})(0.010\ \Omega) = 0.10\ \text{V}$

$$12.0\ \text{V} - 0.10\ \text{V} = 11.9\ \text{V}$$

(b) $V = Ir = (100.0\ \text{A})(0.010\ \Omega) = 1.0\ \text{V}$

$$12.0\ \text{V} - 1.0\ \text{V} = 11.0\ \text{V}$$

To car's electrical system
(ignition, lights, radio, etc.)

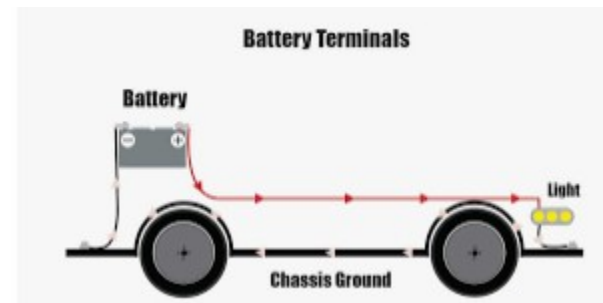
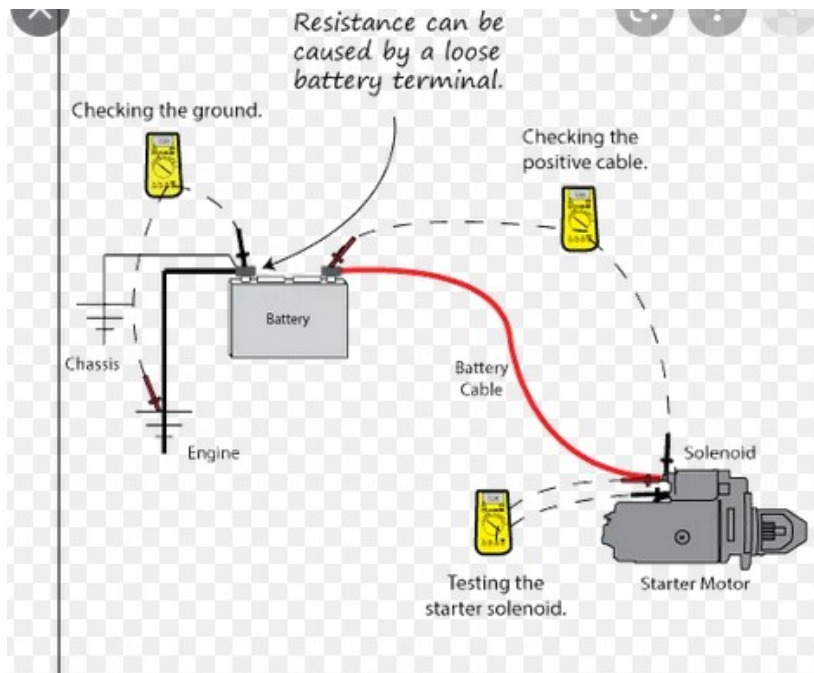


Hard Starting

Corrosion has increased the resistance at your car's battery terminal to $0.03\ \Omega$. If the starter draws a current of 130 A , what's the voltage across the starter?



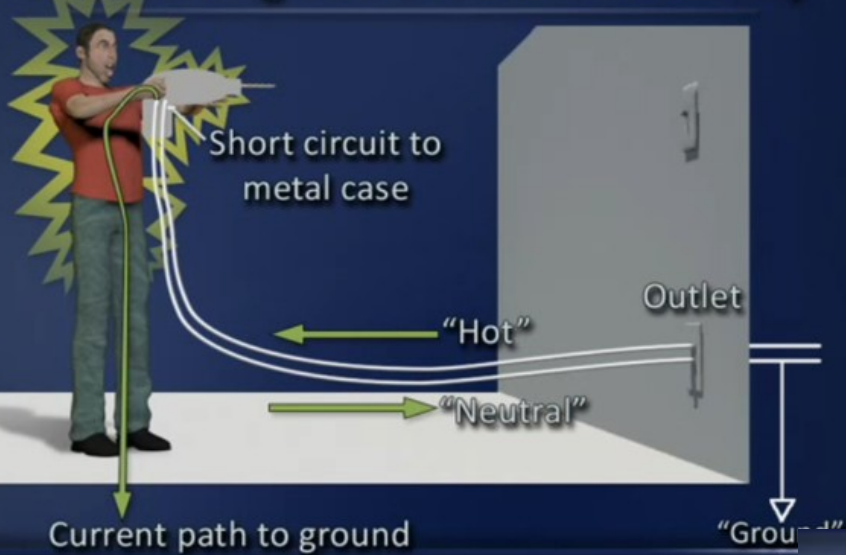
Suppose that the 12V battery provide voltage to the starter and to the resistance. Find V across the resistance then find how much is left for the starter.



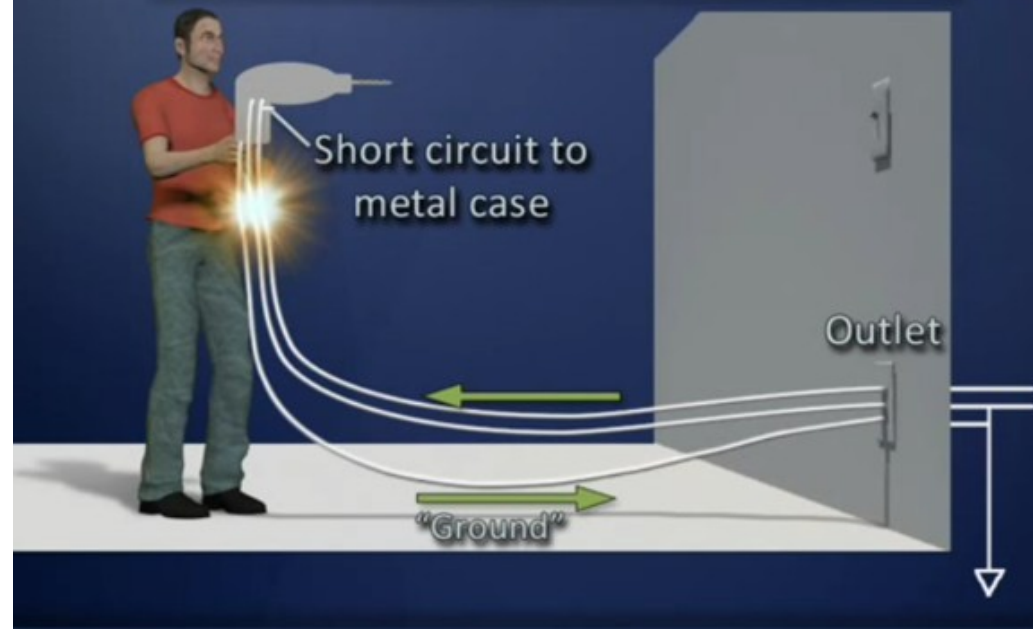
Electrical Safety

Current (mA)	Effect
0.5 - 2	Barely felt
10 - 15	Muscles contract involuntarily
15 - 100	Severe shock; breathing difficult
100 - 200	Heart fibrillation; death in minutes
> 200	Cardiac arrest; burns, breathing stops

Grounding for Electrical Safety

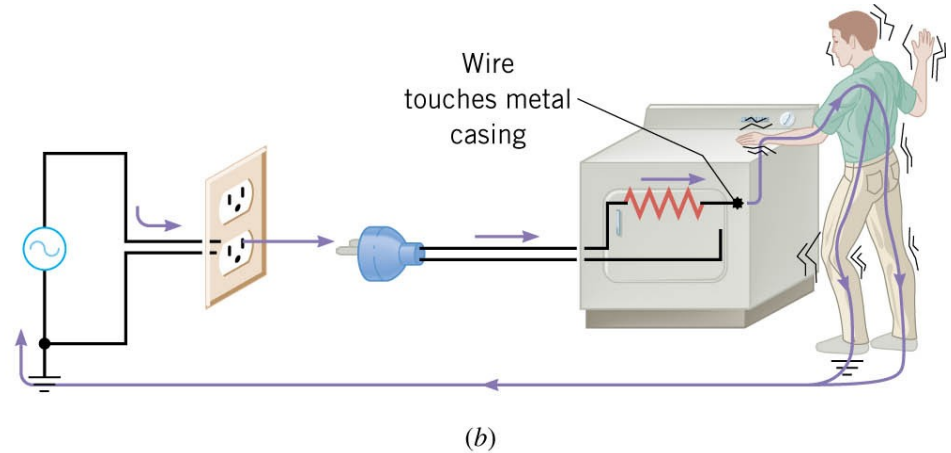
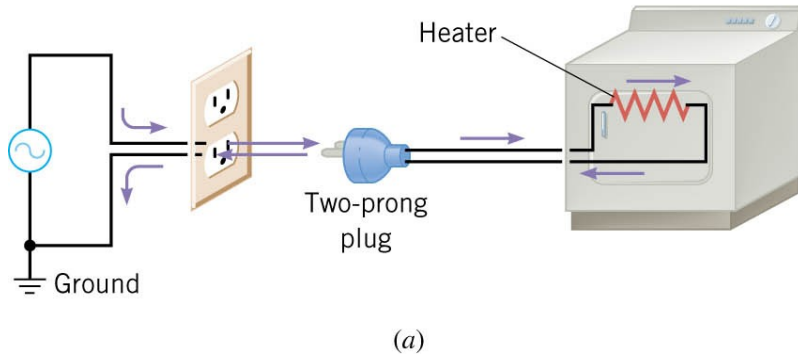


Grounding for Electrical Safety



20.14 Safety and the Physiological Effects of Current

To reduce the danger inherent in using circuits, proper **electrical grounding** is necessary.



20.3 *Resistance and Resistivity*

For a wide range of materials, the resistance of a piece of material of length L and cross-sectional area A is

$$R = \rho \frac{L}{A}$$



resistivity in units of ohm·meter

20.3 Resistance and Resistivity

Table 20.1 Resistivities^a of Various Materials

Material	Resistivity ρ ($\Omega \cdot \text{m}$)	Material	Resistivity ρ ($\Omega \cdot \text{m}$)
Conductors		Semiconductors	
Aluminum	2.82×10^{-8}	Carbon	3.5×10^{-5}
Copper	1.72×10^{-8}	Germanium	0.5^b
Gold	2.44×10^{-8}	Silicon	$20\text{--}2300^b$
Iron	9.7×10^{-8}	Insulators	
Mercury	95.8×10^{-8}	Mica	$10^{11}\text{--}10^{15}$
Nichrome (alloy)	100×10^{-8}	Rubber (hard)	$10^{13}\text{--}10^{16}$
Silver	1.59×10^{-8}	Teflon	10^{16}
Tungsten	5.6×10^{-8}	Wood (maple)	3×10^{10}

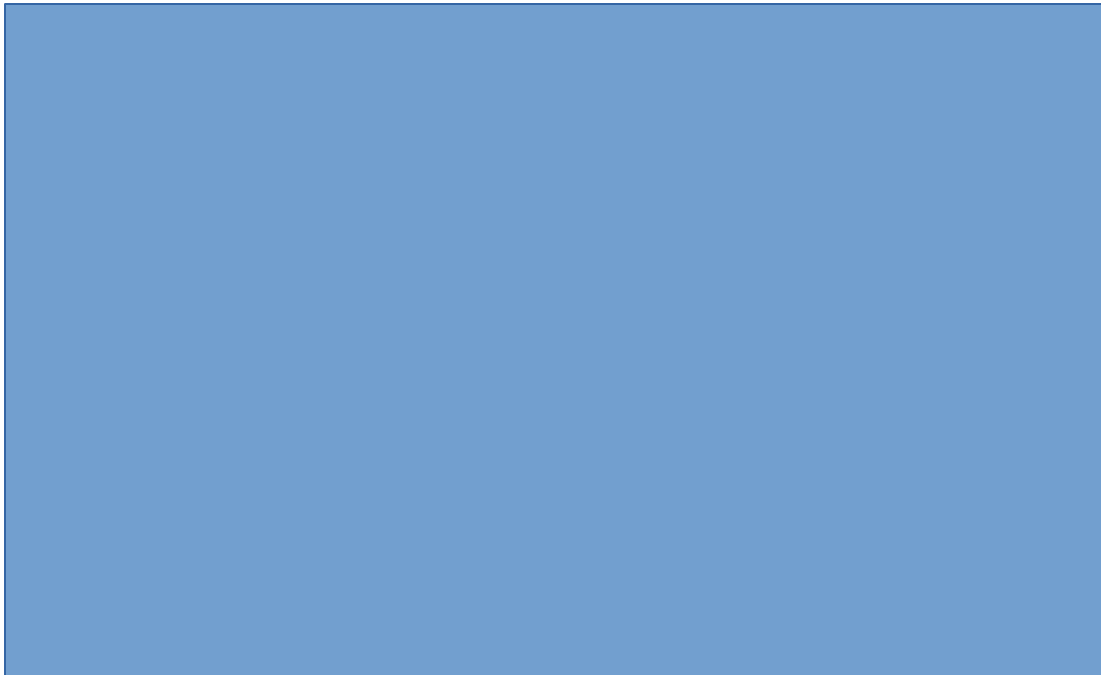
^a The values pertain to temperatures near 20 °C.

^b Depending on purity.

$$R = \rho \frac{L}{A}$$

Example 3 Longer Extension Cords

The instructions for an electric lawn mower suggest that a 20-gauge extension cord can be used for distances up to 35 m, but a thicker 16-gauge cord should be used for longer distances. The cross sectional area of a 20-gauge wire is $5.2 \times 10^{-7} \cdot \text{m}^2$, while that of a 16-gauge wire is $13 \times 10^{-7} \cdot \text{m}^2$. Determine the resistance of (a) 35 m of 20-gauge copper wire and (b) 75 m of 16-gauge copper wire.



Example 3 Longer Extension Cords

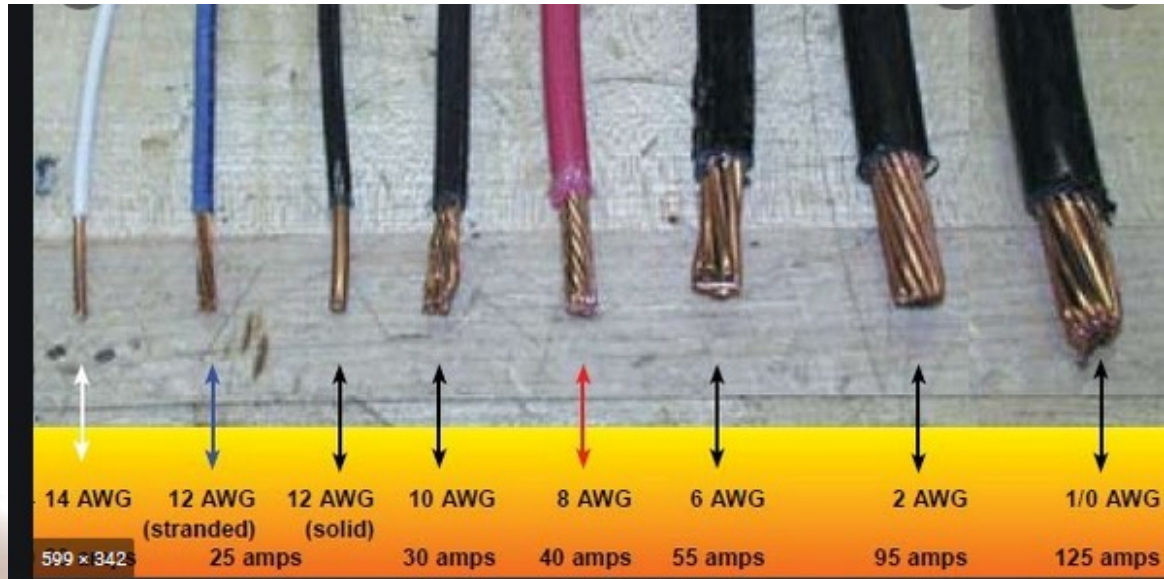
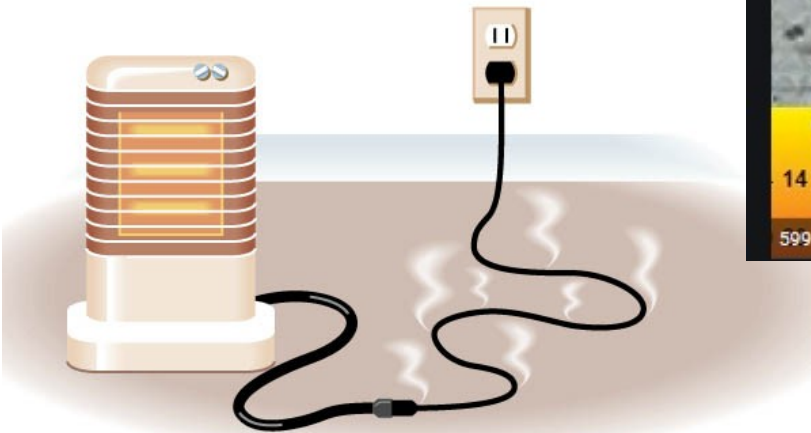
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$$(a) \quad R = \rho \frac{L}{A} = \frac{(1.72 \times 10^{-8} \Omega \cdot \text{m})(35 \text{ m})}{5.2 \times 10^{-7} \text{ m}^2} = 1.2 \Omega$$

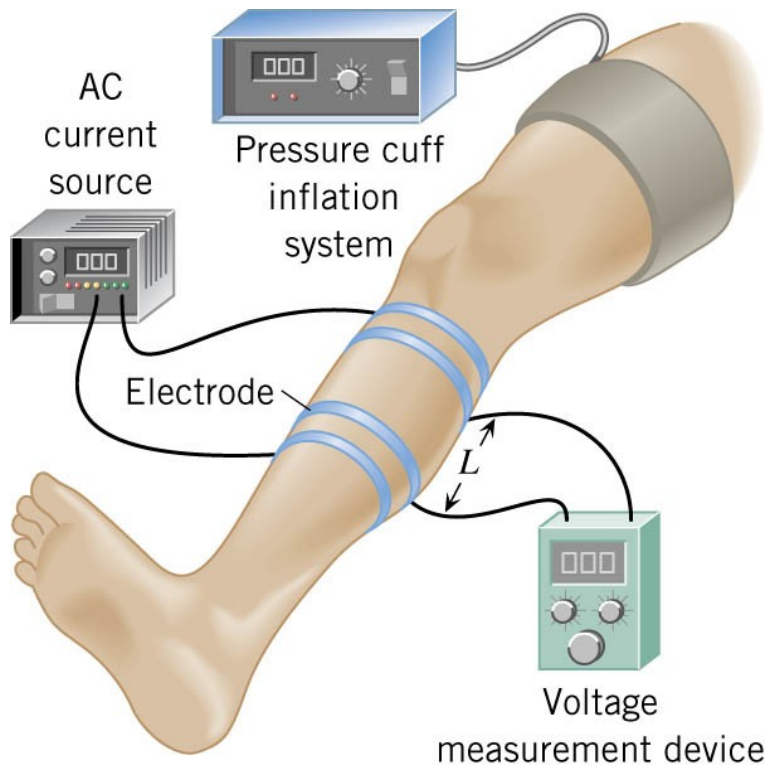
$$(b) \quad R = \rho \frac{L}{A} = \frac{(1.72 \times 10^{-8} \Omega \cdot \text{m})(75 \text{ m})}{13 \times 10^{-7} \text{ m}^2} = 0.99 \Omega$$

Conceptual Example 7 Extension Cords and a Potential Fire Hazard

During the winter, many people use portable electric space heaters to keep warm. Sometimes, however, the heater must be located far from a 120-V wall receptacle, so an extension cord must be used. However, manufacturers often warn against using an extension cord. If one must be used, they recommend a certain wire gauge, or smaller. Why the warning, and why are smaller-gauge wires better than larger-gauge wires?



20.3 Resistance and Resistivity



$$R = \rho \frac{L}{A} = \rho \frac{L}{V_{\text{calf}} / L} = \rho \frac{L^2}{V_{\text{calf}}}$$

Impedance plethysmography is a method of determining changing tissue volumes in the body, based on the measurement of electric impedance at the body surface.

$$\rho = \rho_o \left[1 + \alpha (T - T_o) \right]$$

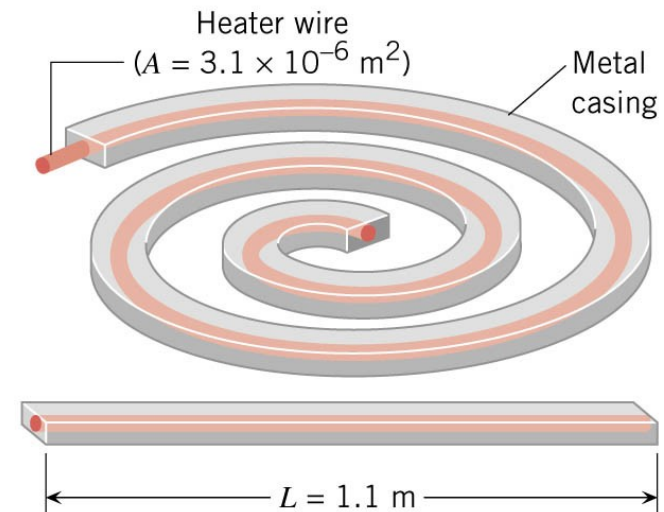
temperature coefficient
of resistivity. True for conductors.

Resistance increases with temperature.
In a semi-conductor, the resistance
Decreases with temperature. We
Use semi-conductor to measure temp.

$$R = R_o \left[1 + \alpha (T - T_o) \right]$$



(a)



(b)

ELECTRIC POWER

20.4 *Electric Power*

When there is current in a circuit as a result of a voltage, the electric power delivered to the circuit is:

$$P = IV$$

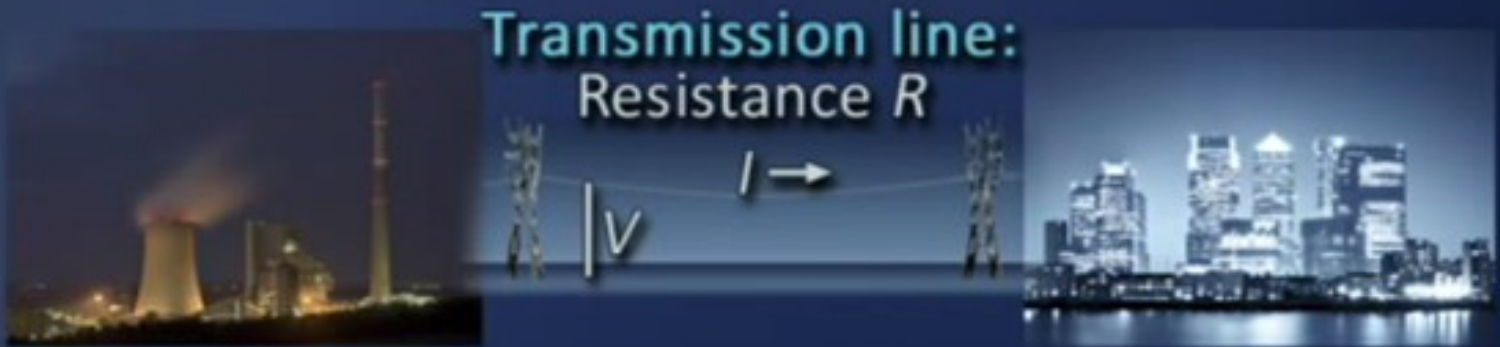
SI Unit of Power: watt (W)

Many electrical devices are essentially resistors:

$$P = I (IR) = I^2 R$$

$$P = \left(\frac{V}{R} \right) V = \frac{V^2}{R}$$

Electric Power Transmission



Power transmitted to city: $P = IV$

Large I , small V ...

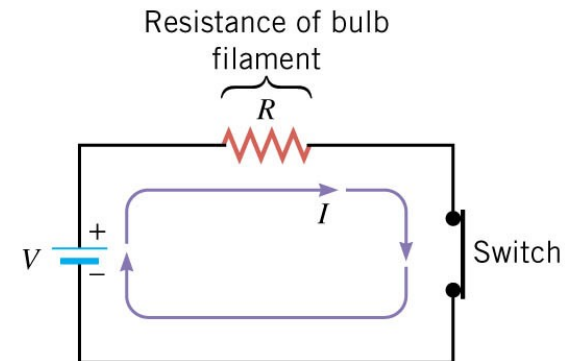
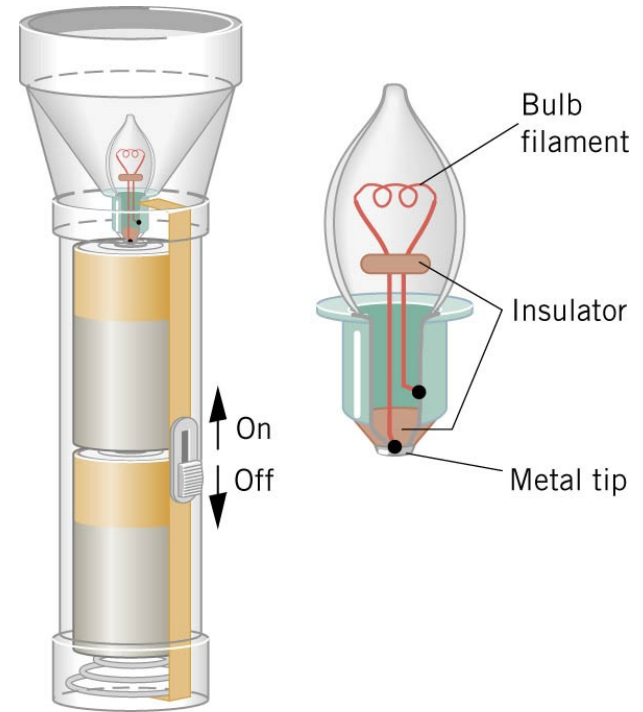
...or small I , large V

Power lost in transmission line: $P = I^2 R$

20.4 Electric Power

Example 5 The Power and Energy Used in a Flashlight

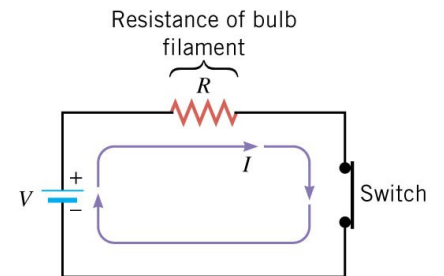
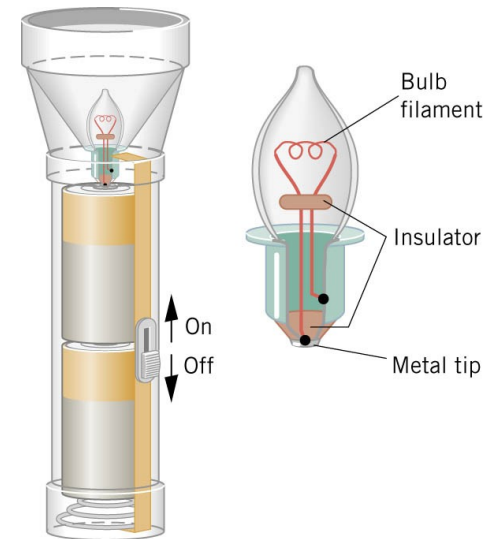
In the flashlight, the current is 0.40 A and the voltage is 3.0 V . Find (a) the power delivered to the bulb and (b) the energy dissipated in the bulb in 5.5 minutes of operation.



20.4 Electric Power

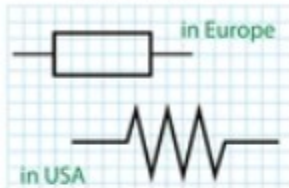
(a) $P = IV = (0.40 \text{ A})(3.0 \text{ V}) = 1.2 \text{ W}$

(b) $E = Pt = (1.2 \text{ W})(330 \text{ s}) = 4.0 \times 10^2 \text{ J}$

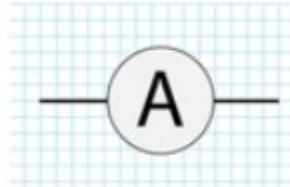


ELECTRICAL CIRCUIT SYMBOLS

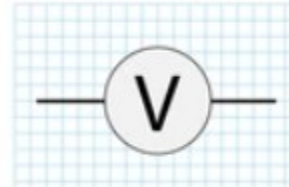
Resistor



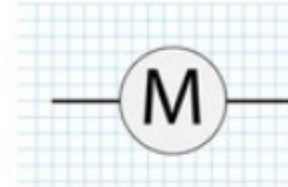
Ammeter



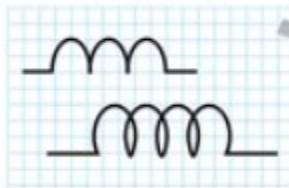
Voltmeter



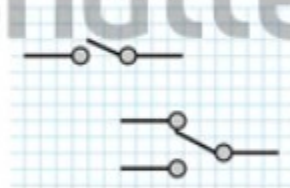
Motor



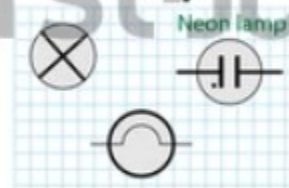
Inductor



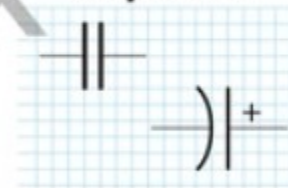
Switch



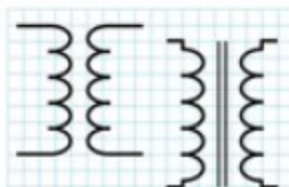
Lamp



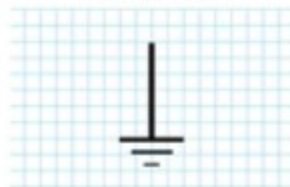
Capacitor



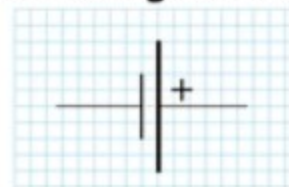
Transformer



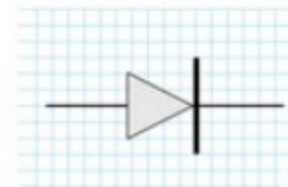
Ground



DC voltage source



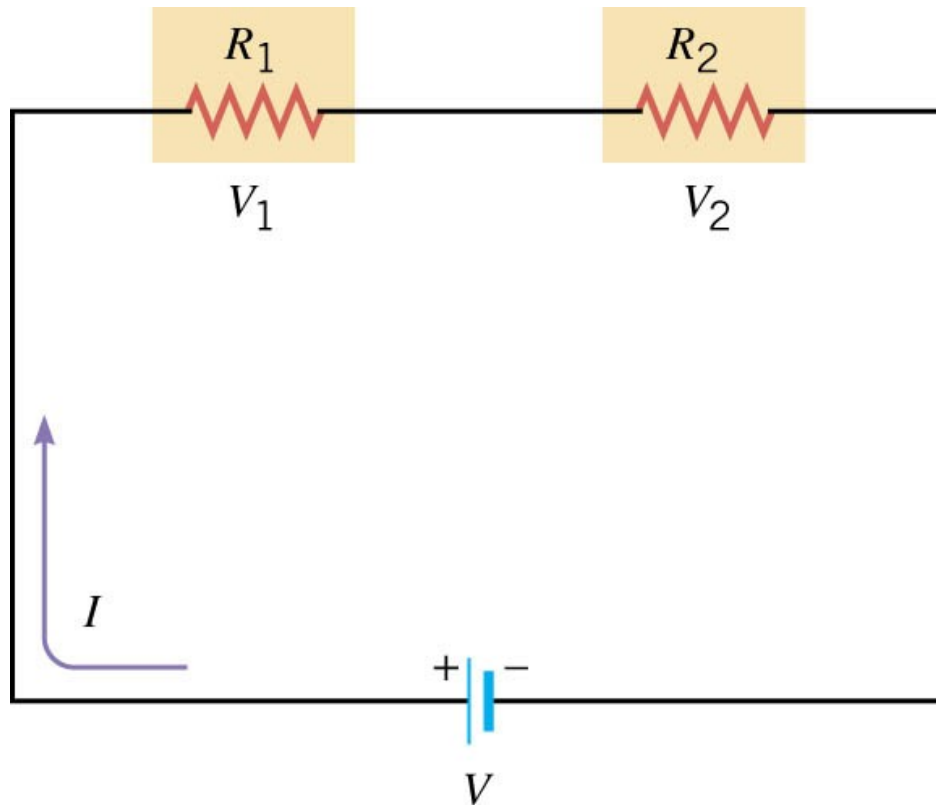
Diode



20.6 Series Wiring

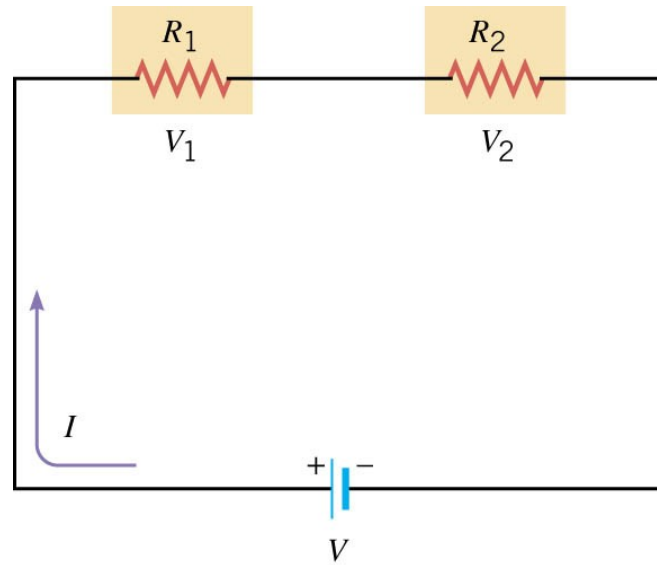
There are many circuits in which more than one device is connected to a voltage source.

Series wiring means that the devices are connected in such a way that there is the same electric current through each device.



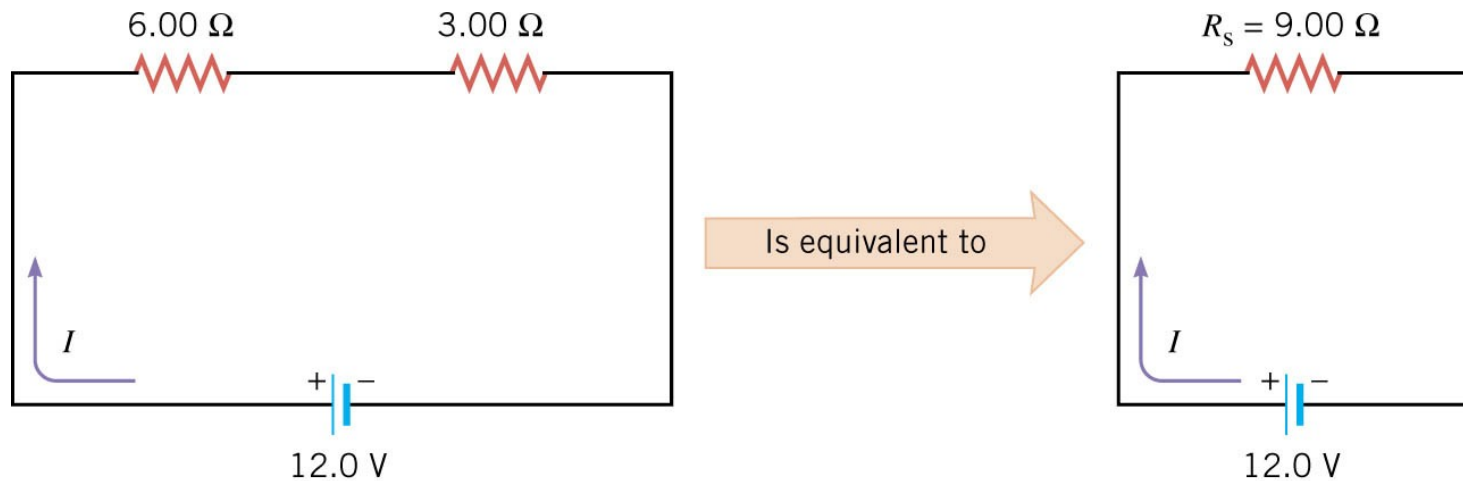
20.6 Series Wiring

$$V = V_1 + V_2 = IR_1 + IR_2 = I(R_1 + R_2) = IR_S$$



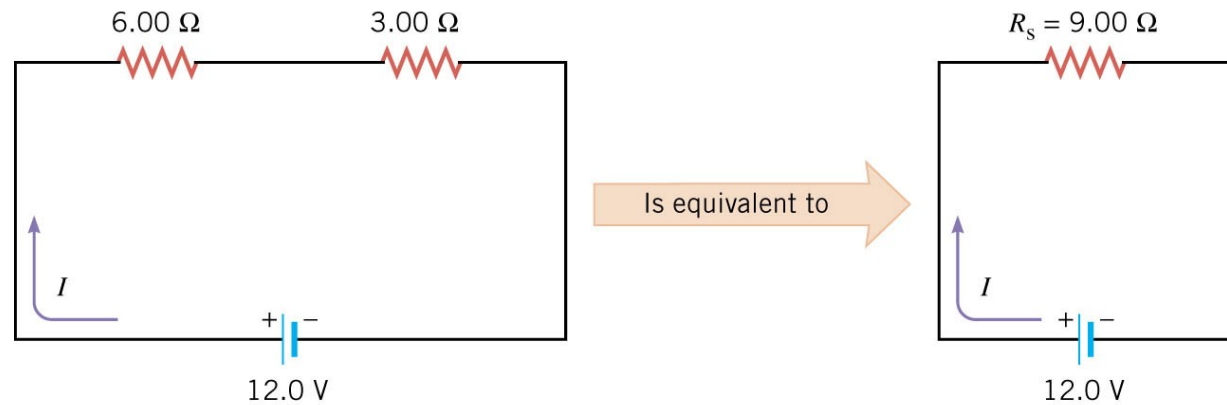
Series resistors

$$R_S = R_1 + R_2 + R_3 + \dots$$



Example 8 Resistors in a Series Circuit

A 6.00 Ω resistor and a 3.00 Ω resistor are connected in series with a 12.0 V battery. Assuming the battery contributes no resistance to the circuit, find (a) the current, (b) the power dissipated in each resistor, and (c) the total power delivered to the resistors by the battery.



(a) $R_s = 6.00 \Omega + 3.00 \Omega = 9.00 \Omega$ $I = \frac{V}{R_s} = \frac{12.0 \text{ V}}{9.00 \Omega} = 1.33 \text{ A}$

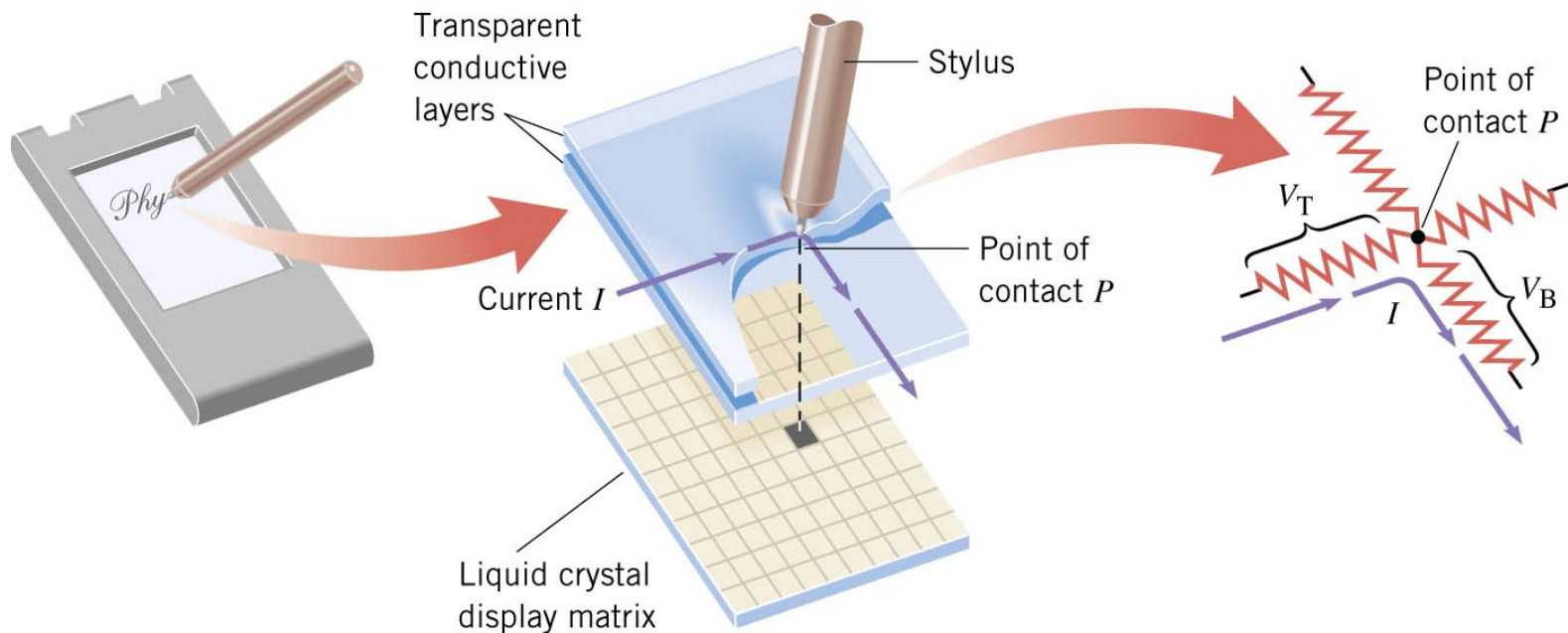
(b) $P = I^2 R = (1.33 \text{ A})^2 (6.00 \Omega) = 10.6 \text{ W}$

$$P = I^2 R = (1.33 \text{ A})^2 (3.00 \Omega) = 5.31 \text{ W}$$

(c) $P = 10.6 \text{ W} + 5.31 \text{ W} = 15.9 \text{ W}$

20.6 Series Wiring

Personal electronic assistants.

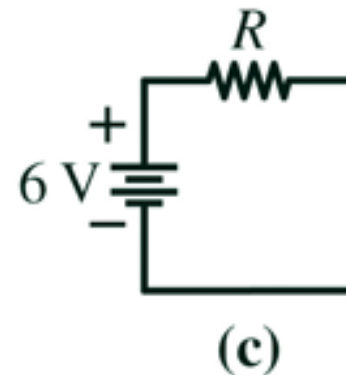
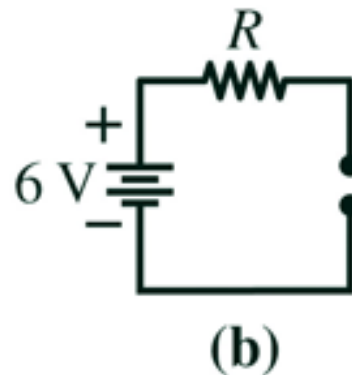
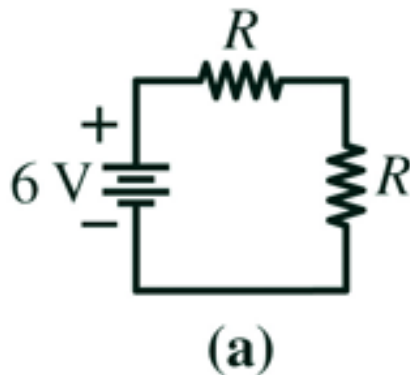


If I is detected, R are the same, Then the voltages V_T and V_B Are computed $\rightarrow$ position in the matrix.

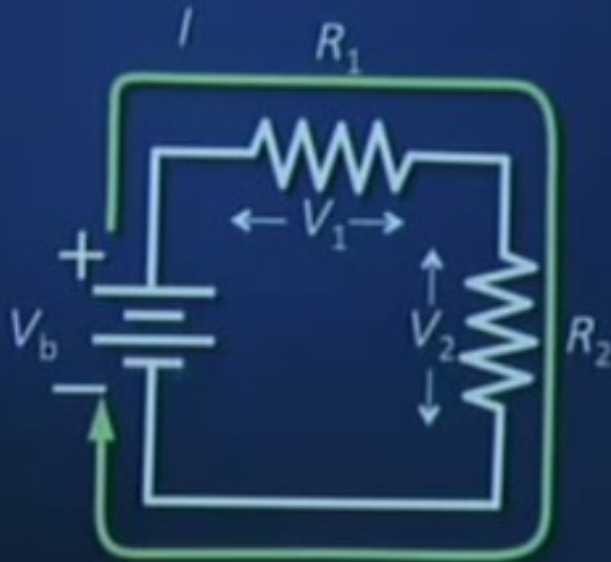
- Rank order the voltages across the identical resistors R at the top of each circuit shown. In (a) the second resistor has the same resistance R , and in (b) the gap is an open circuit (infinite resistance).

- A. (c) > (b) > (a)
- B. (c) > (a) > (b)
- C. (b) > (c) > (a)
- D. (a) > (b) > (c)

Find the voltage across the Top resistor.



Series Resistors: A Voltage Divider



Special cases:

$$R_1 = R_2: V_1 = V_2 = \frac{1}{2}V_b$$

$$R_1 \gg R_2: V_2 \ll V_b$$

$$R_1 \ll R_2: V_2 \simeq V_b$$

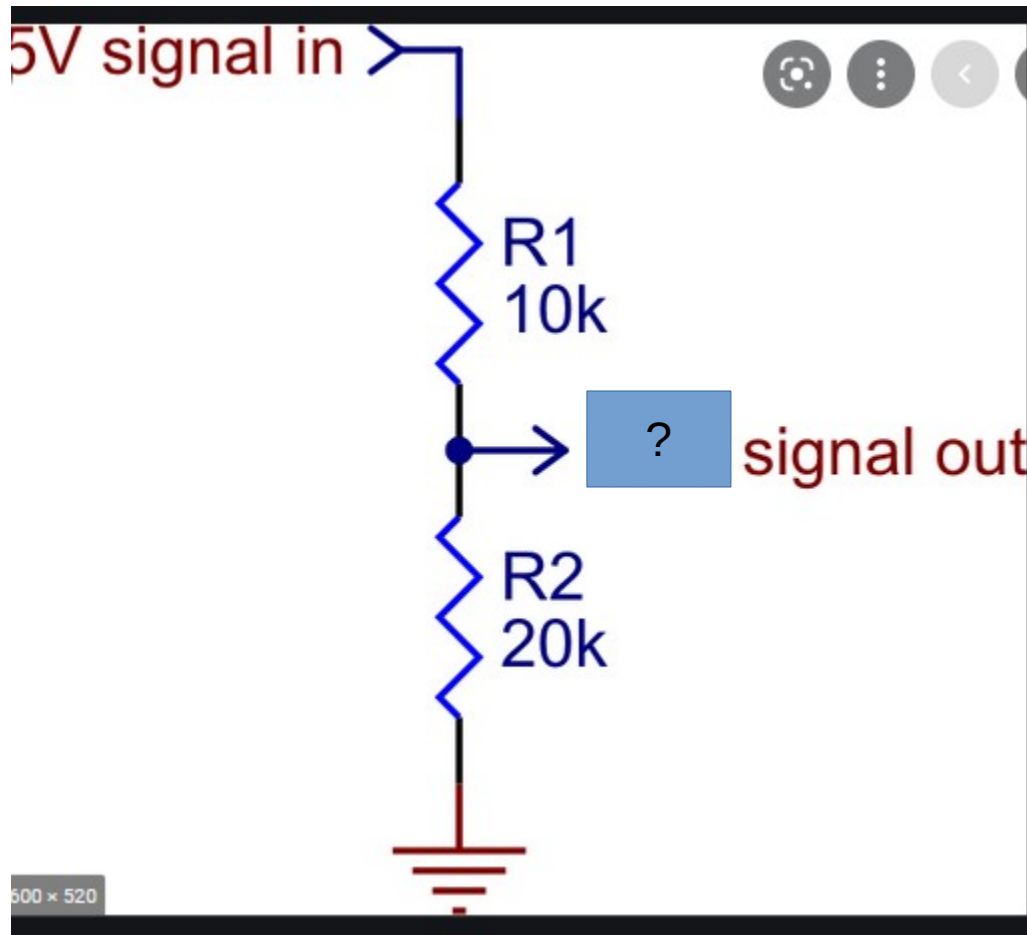
$$V_1 = IR_1 \quad V_2 = IR_2$$

$$I = \frac{V_b}{R_1 + R_2}$$

So:

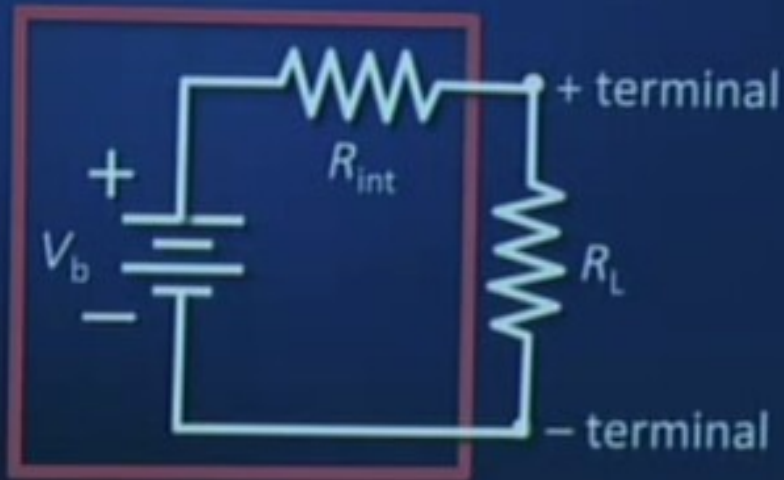
$$V_1 = IR_1 = \left(\frac{R_1}{R_1 + R_2} \right) V_b$$
$$V_2 = IR_2 = \left(\frac{R_2}{R_1 + R_2} \right) V_b$$

Voltage divider fraction



Arduino.
Provides 5V or 3.3V

A Real Battery



Special cases:

$$R_{int} \ll R_L : V_L \simeq V_b$$

$$R_{int} \gg R_L : V_L \ll V_b$$

~~Ideal battery: Voltage between terminals is V_b no matter what~~

Real battery: *Internal resistance R_{int}*

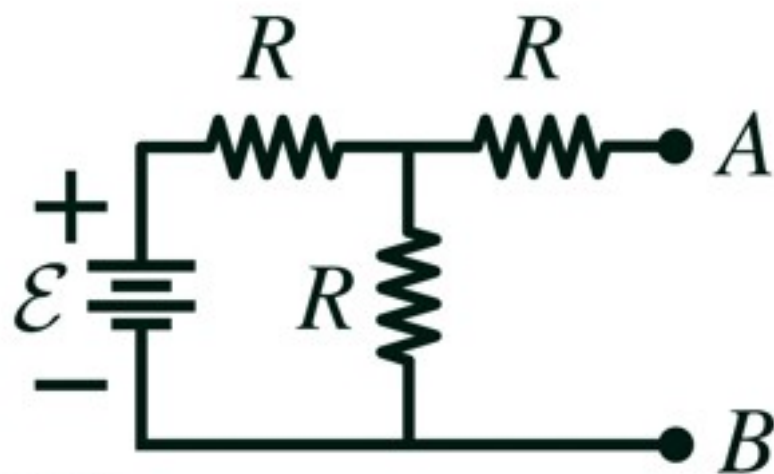
Voltage divider: $V_2 = \left(\frac{R_2}{R_1 + R_2} \right) V_b$

$$\Rightarrow V_L = \left(\frac{R_L}{R_{int} + R_L} \right) V_b$$

Do textbook 25.8, 25.9, 25.10 . A 100W bulb has a smaller resistance than a 50W bulb

- An ideal voltmeter is connected between points A and B in the figure. What value does the voltmeter read?

- A. ε
- B. $\varepsilon/2$
- C. $\varepsilon/3$
- D. $2\varepsilon/3$



2. Wheatstone Bridge

In the circuit at right, R_4 is the unknown resistance; R_1 , R_2 and R_3 are resistors of known resistance where the resistance of R_3 is adjustable.

How to determine the resistance of the unknown resistor, R_4 ?

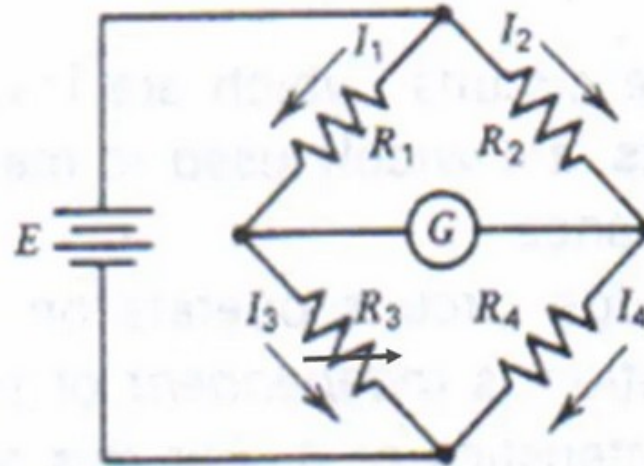
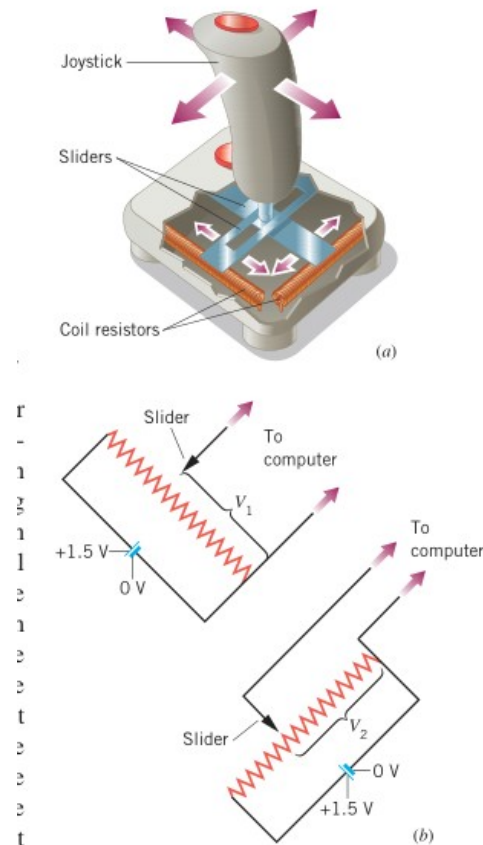


Figure 5.1: Wheatstone Bridge Circuit

“The resistances of the other three are **adjusted** and **balanced** until the current passing through the galvanometer **decreases to zero**”.

When the current through the G (includes a resistance) is 0 then $V_3 = V_4$ and $V_1 = V_2$. You can use the voltage divider equation $V_3/E = R_3/(R_1 + R_3)$ and $V_4/E = R_4/(R_2 + R_4)$ so $R_3/(R_1 + R_3) = R_4/(R_2 + R_4)$. Flip them → $R_1/R_3 = R_2/R_4$ → solve for R_4



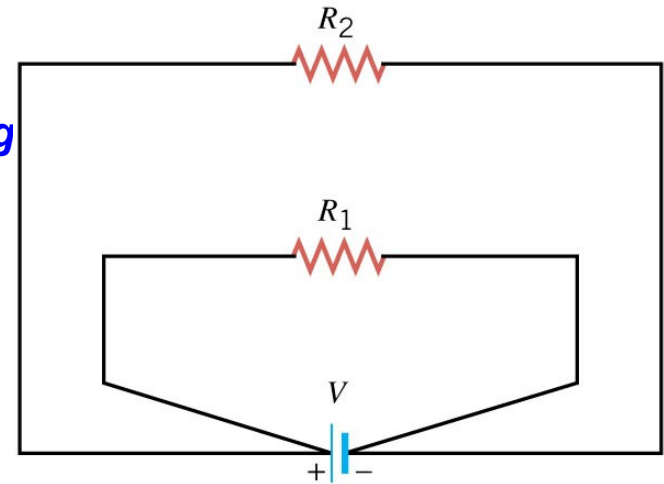
The physics of a joystick. The joystick, found in computer games, also takes advantage of resistors connected in series. A joystick contains two straight coils of resistance wire that are oriented at 90° to each other (see Figure 20.17a). When the joystick is moved, it repositions the metallic slider on each of the coils. As part *b* of the drawing illustrates, each coil is connected across a 1.5-V battery.* Because one end of a coil is at 1.5 V and the other at 0 V, the voltage at the location of the slider is somewhere between these values; the voltage of the left slider in the drawing is labeled V_1 and the voltage of the right slider is V_2 . The slider voltages are sent via wires to a computer, which translates them into positional data. In effect, the slider divides each resistance coil into two smaller resistance coils wired in series, and allows the voltage at the point where they are joined together to be detected.

20.7 Parallel Wiring

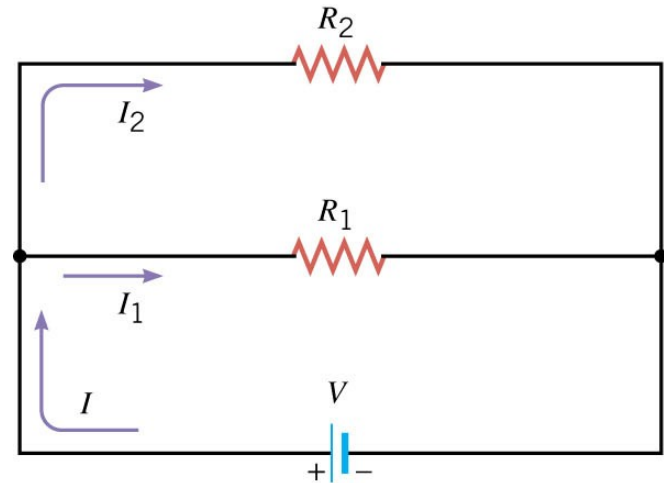
Parallel wiring means that the devices are connected in such a way that the same voltage is applied across each device.

When two resistors are connected in parallel, each receives current from the battery as if the other was not present.

Therefore the two resistors connected in parallel draw more current than does either resistor alone.



(a)



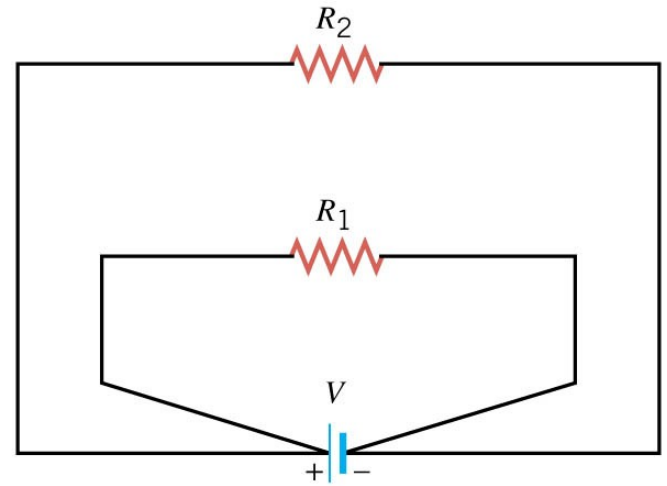
(b)

20.7 Parallel Wiring

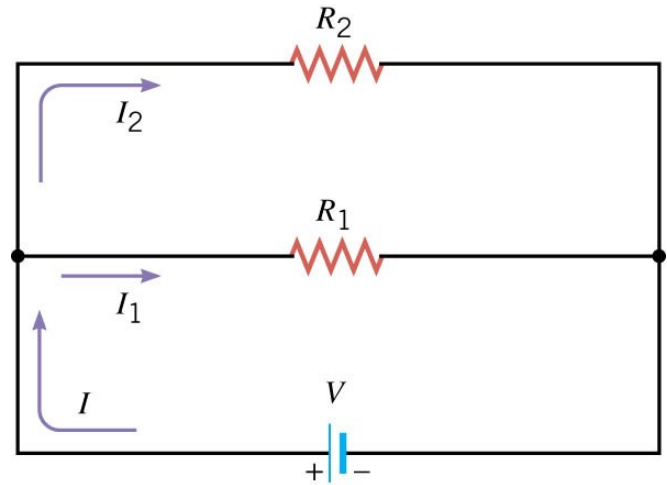
$$I = I_1 + I_2 = \frac{V}{R_1} + \frac{V}{R_2} = V \left(\frac{1}{R_1} + \frac{1}{R_2} \right) = V \left(\frac{1}{R_P} \right)$$

parallel resistors

$$\frac{1}{R_P} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \dots$$

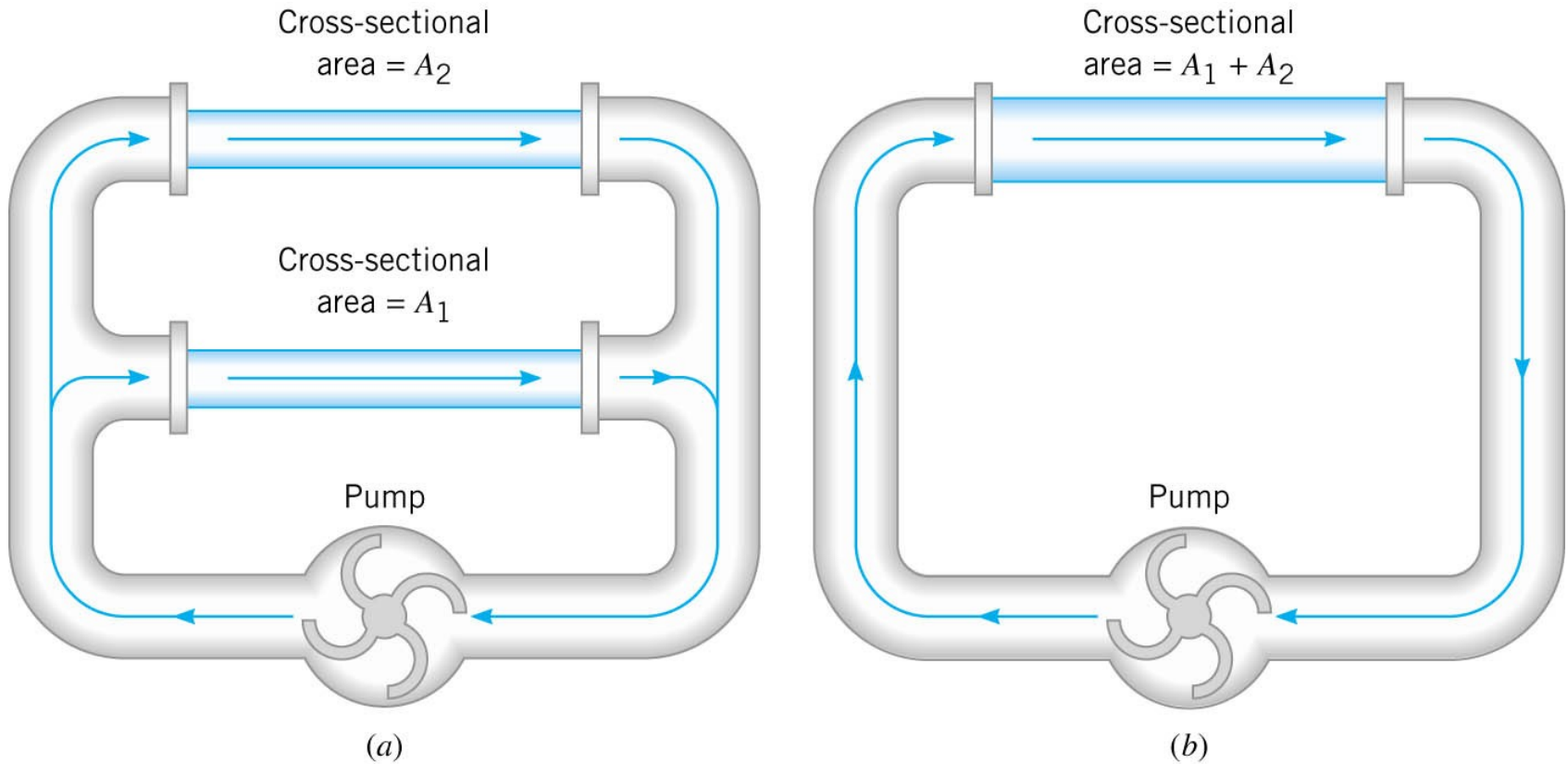


(a)



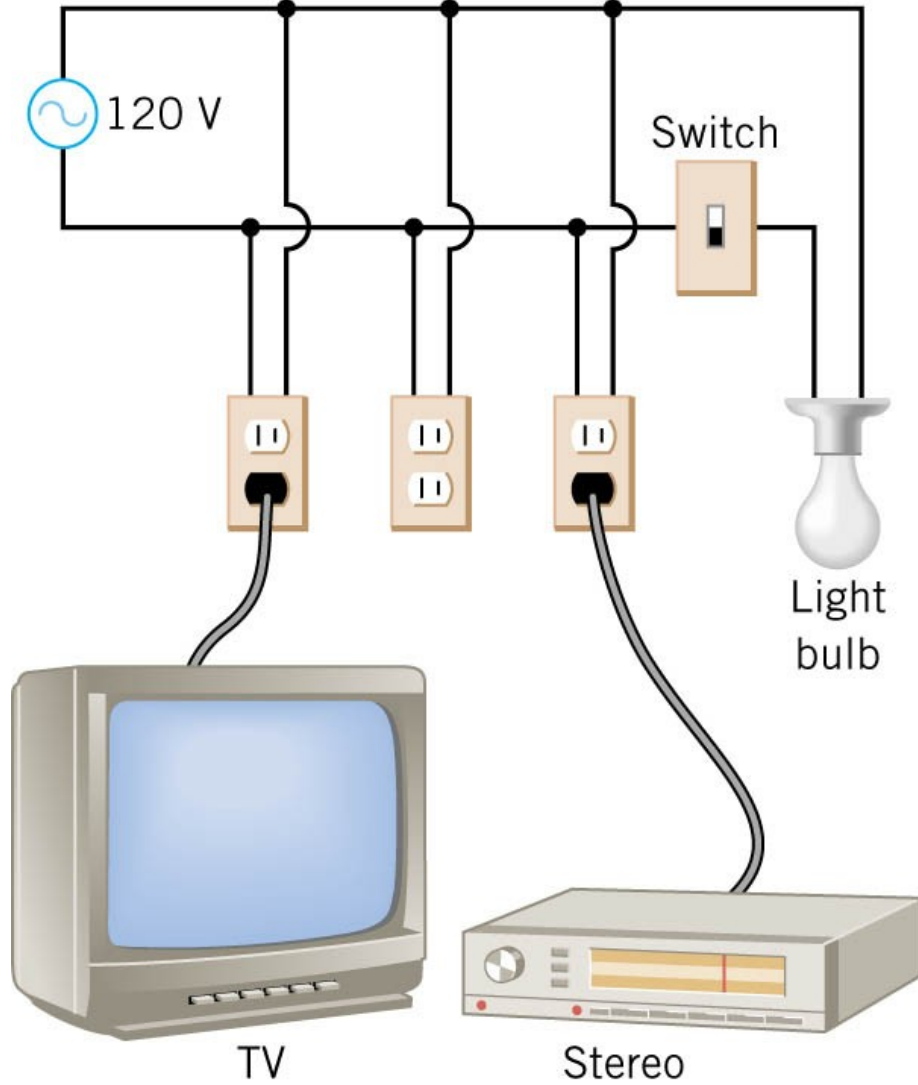
(b)

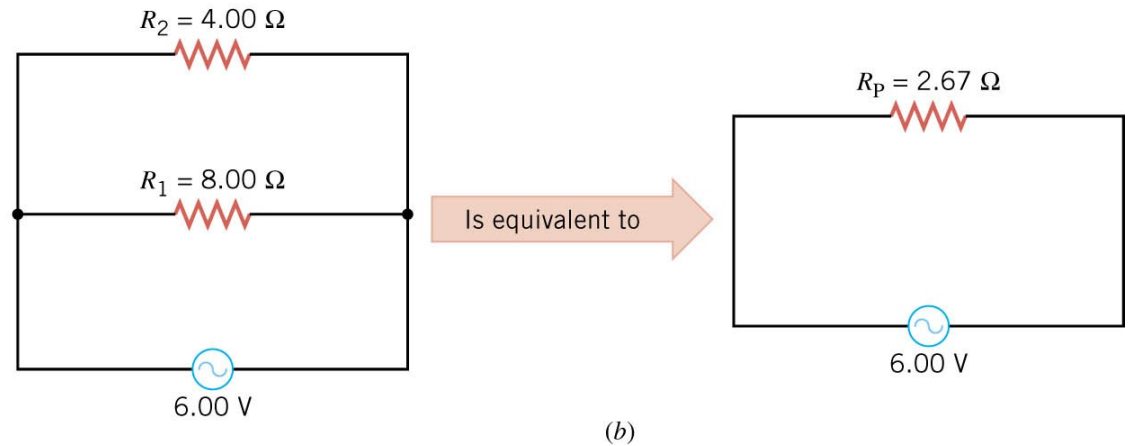
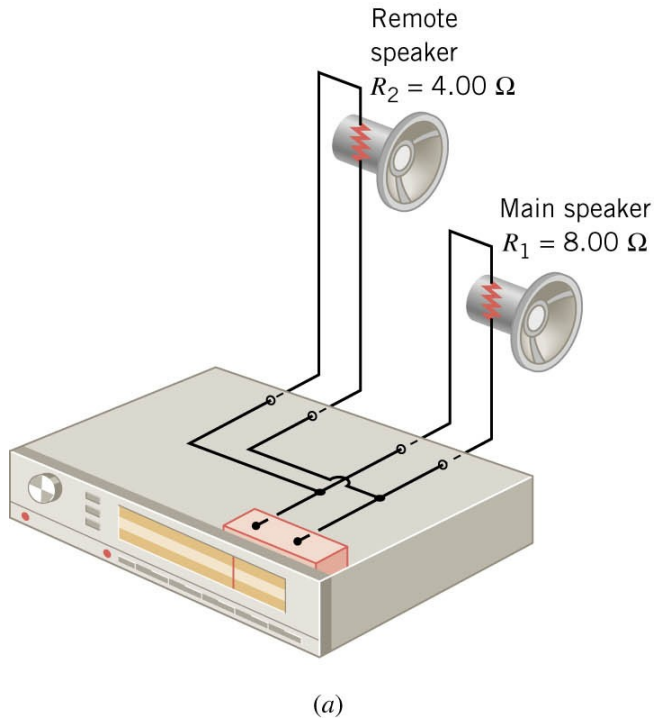
20.7 Parallel Wiring



The two parallel pipe sections are equivalent to a single pipe of the same length and same total cross sectional area.

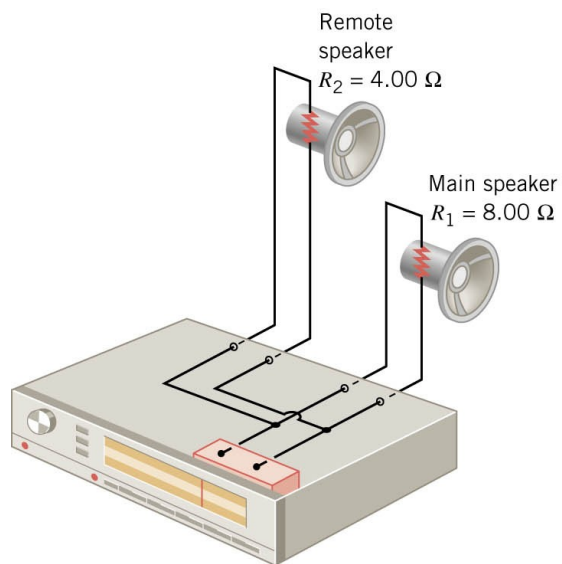
20.7 Parallel Wiring



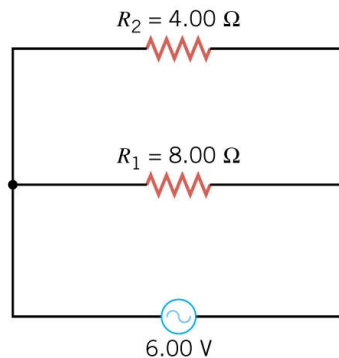


Example 10 Main and Remote Stereo Speakers

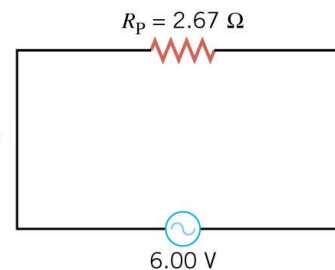
Most receivers allow the user to connect to “remote” speakers in addition to the main speakers. At the instant represented in the picture, the voltage across the speakers is 6.00 V. Determine (a) the equivalent resistance of the two speakers, (b) the total current supplied by the receiver, (c) the current in each speaker, and (d) the power dissipated in each speaker.



(a)



Is equivalent to

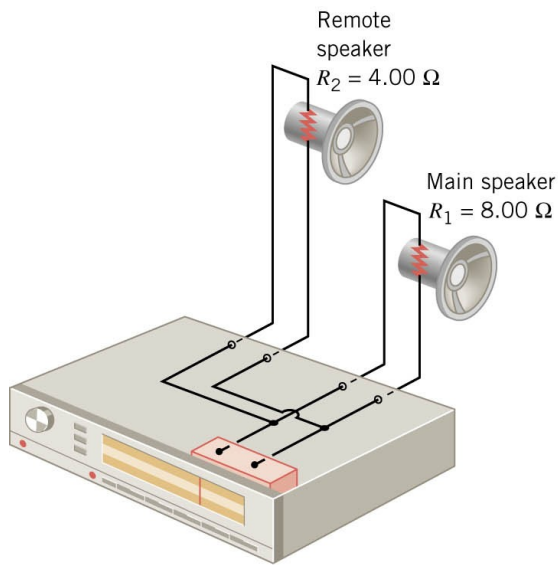


(b)

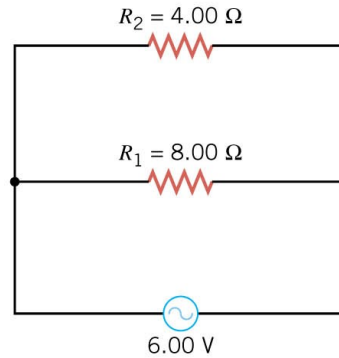
$$(a) \quad \frac{1}{R_p} = \frac{1}{8.00 \, \Omega} + \frac{1}{4.00 \, \Omega} = \frac{3}{8.00 \, \Omega}$$

$$R_p = 2.67 \, \Omega$$

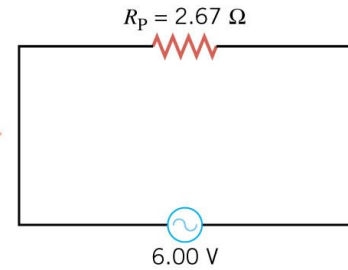
$$(b) \quad I_{\text{rms}} = \frac{V_{\text{rms}}}{R_p} = \frac{6.00 \, \text{V}}{2.67 \, \Omega} = 2.25 \, \text{A}$$



(a)



Is equivalent to



(b)

(c)
$$I_{\text{rms}} = \frac{V_{\text{rms}}}{R} = \frac{6.00 \text{ V}}{8.00 \Omega} = 0.750 \text{ A}$$

$$I_{\text{rms}} = \frac{V_{\text{rms}}}{R} = \frac{6.00 \text{ V}}{4.00 \Omega} = 1.50 \text{ A}$$

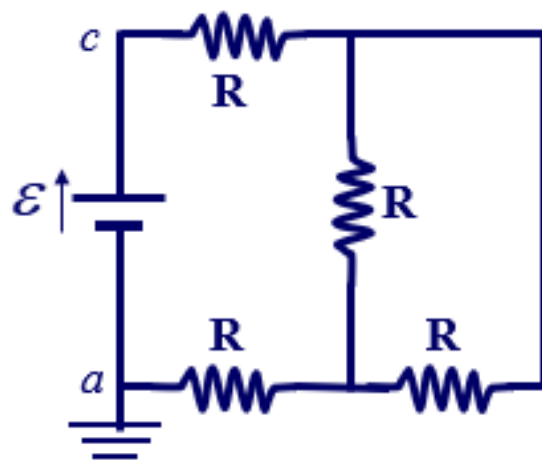
(d)
$$\bar{P} = I_{\text{rms}} V_{\text{rms}} = (0.750 \text{ A})(6.00 \text{ V}) = 4.50 \text{ W}$$

$$\bar{P} = I_{\text{rms}} V_{\text{rms}} = (1.50 \text{ A})(6.00 \text{ V}) = 9.00 \text{ W}$$

Resistors in series and parallel

2. Four resistors are connected as shown in figure. Find the equivalent resistance between points a and c .

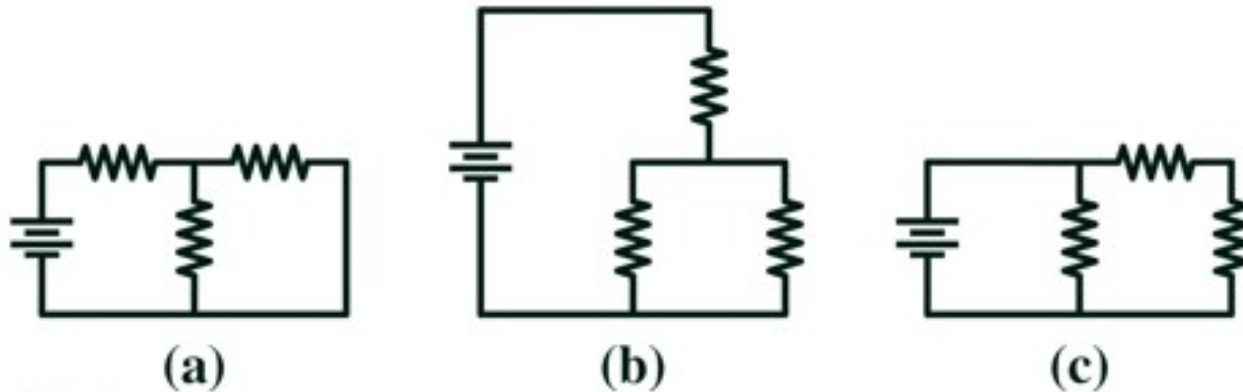
- A. $4 R$.
- B. $3 R$.
- C. $2.5 R$.
- D. $0.4 R$.
- E. Cannot determine from information given.



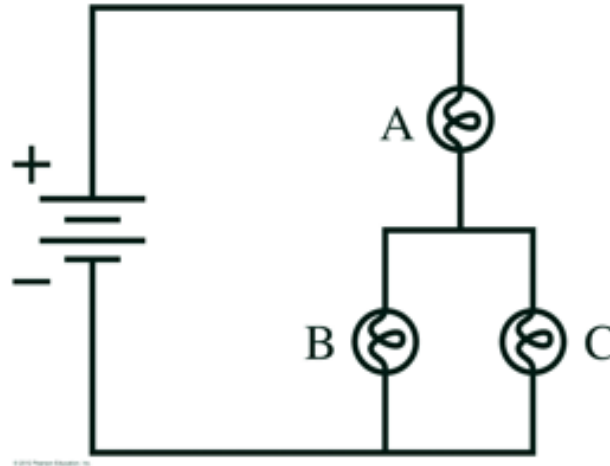
- The figure shows three circuits, each with three identical resistors. Which two are electrically equivalent?

- A. Circuits (a) and (c)
- B. Circuits (b) and (c)
- C. Circuits (b) and (a)

Also . Draw each circuit engineering way



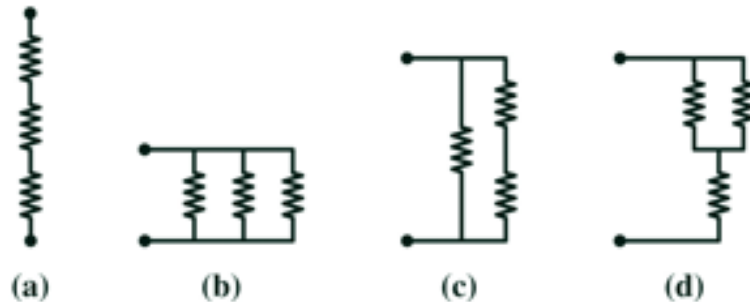
- The figure shows a circuit with three identical lightbulbs and a battery. What happens to each of the other two bulbs if you remove bulb C?
- A. Bulb A brightens; bulb B dims.
B. Bulb B brightens; bulb A dims.
C. Bulbs A and B both dim.

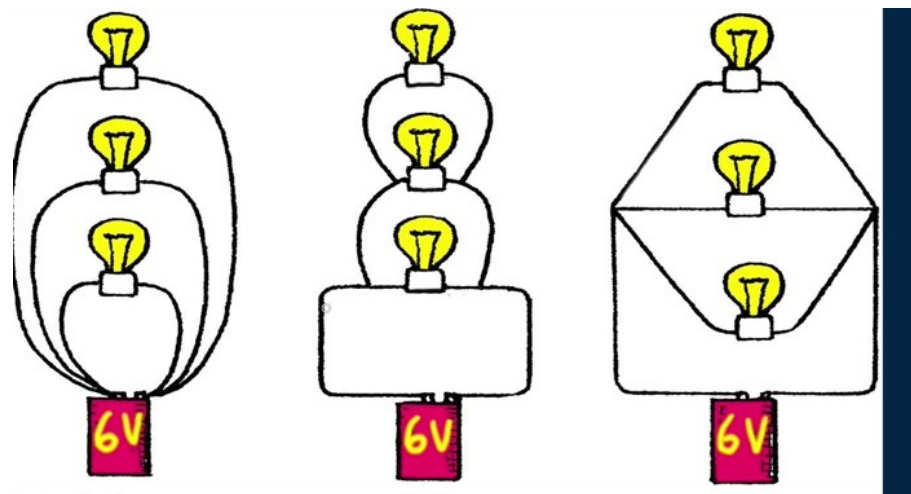
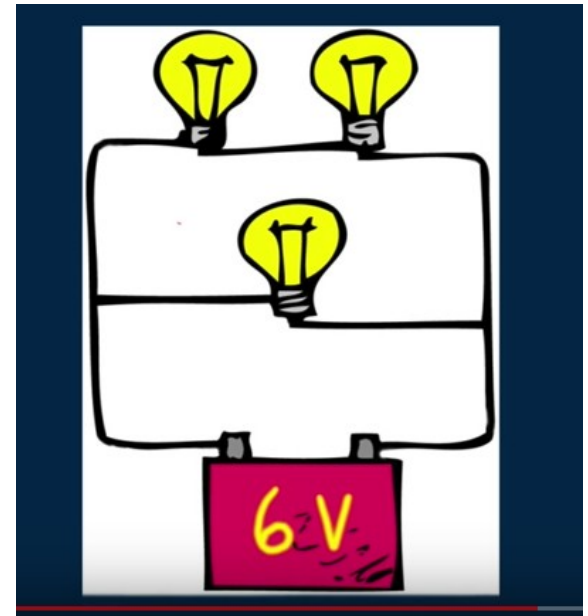
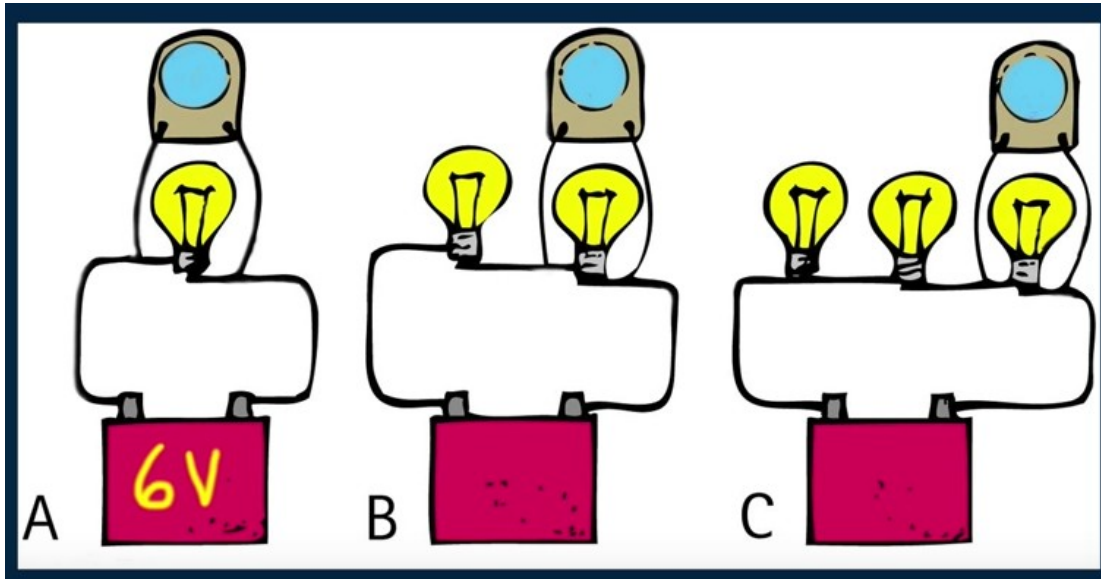


The brightness depends on the power $= R I^2$
Take $V = 1 \text{ V}$ and $R = 1 \text{ ohms}$ if it helps. R is the same for all the loads.
So power only depends on I^2 or I

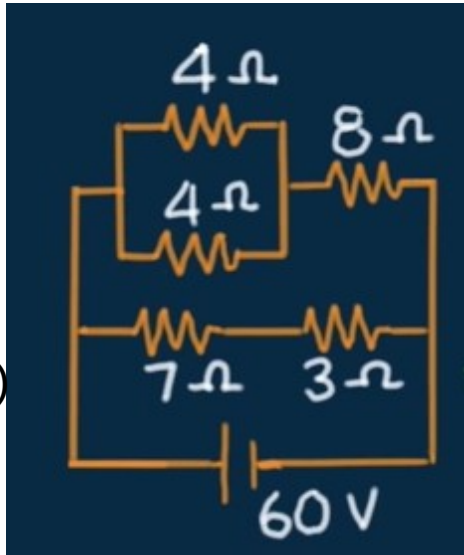
- The figure shows all four possible combinations of three identical resistors. How are they ranked in order of decreasing resistance?

- A. $(b) > (d) > (c) > (a)$
- B. $(a) > (c) > (d) > (b)$
- C. $(b) > (c) > (d) > (a)$
- D. $(a) > (d) > (c) > (b)$

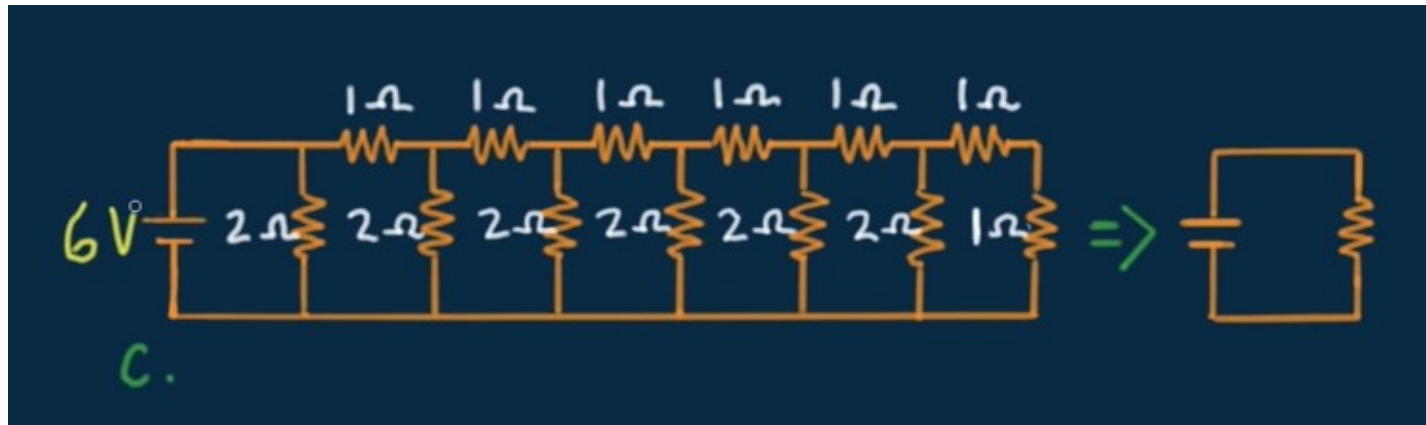
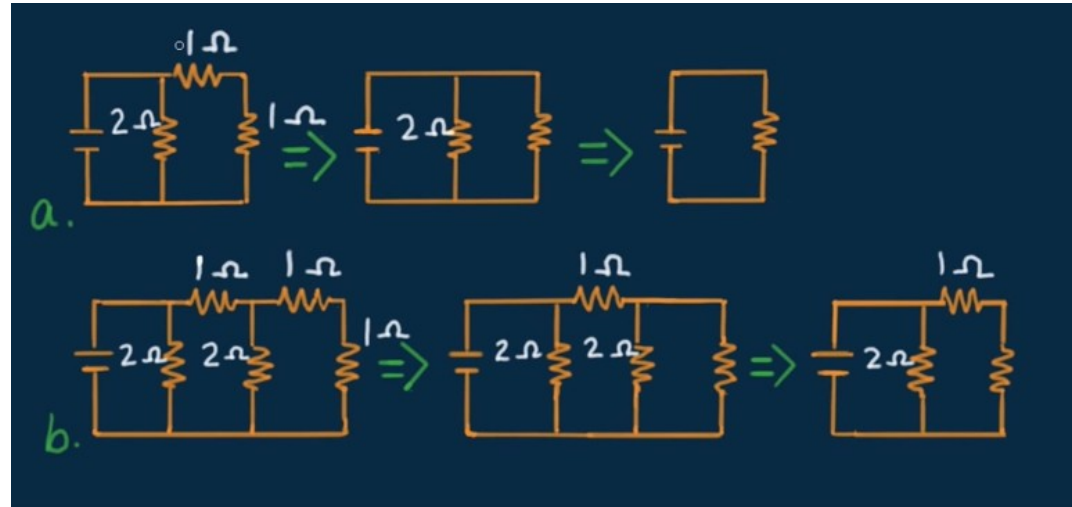




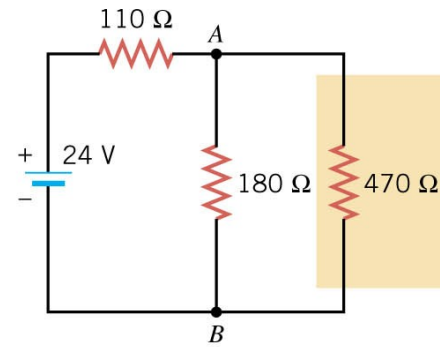
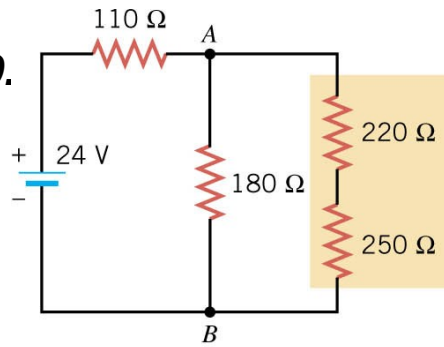
2)



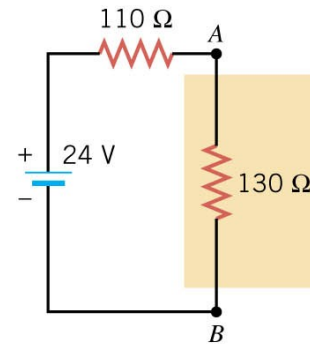
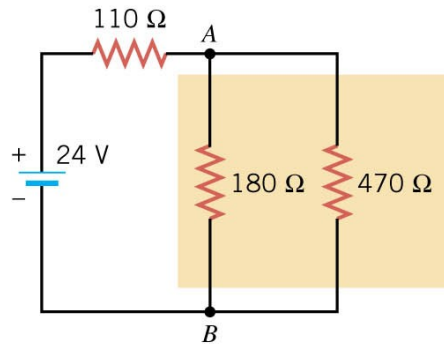
3)



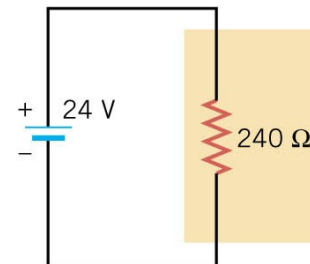
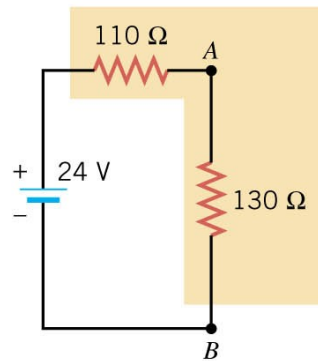
20.



(a)



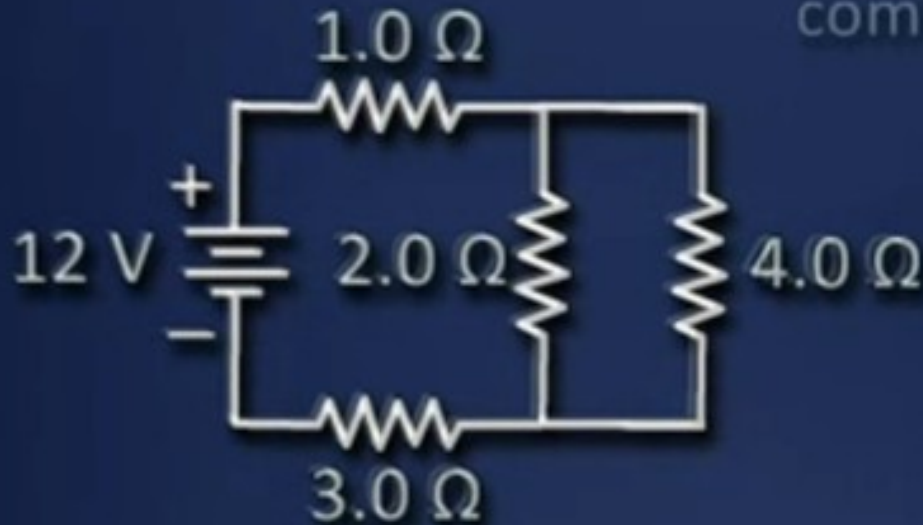
(b)

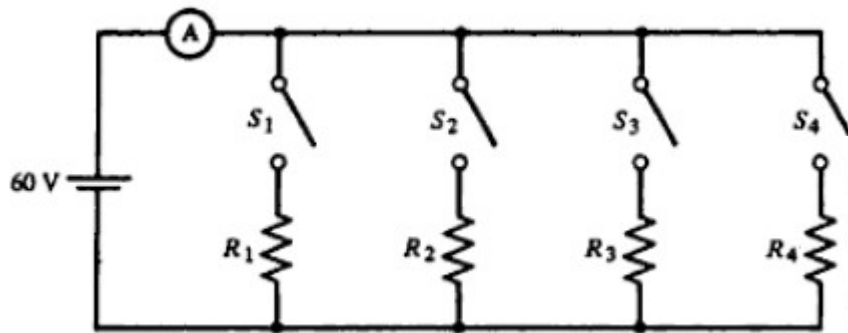


(c)

Analyzing Complex Circuits

- Identify series or parallel combination

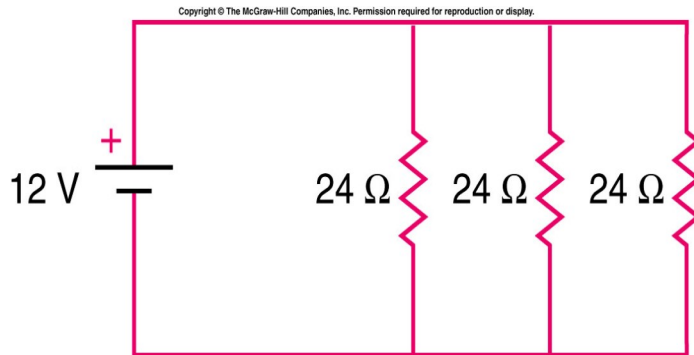




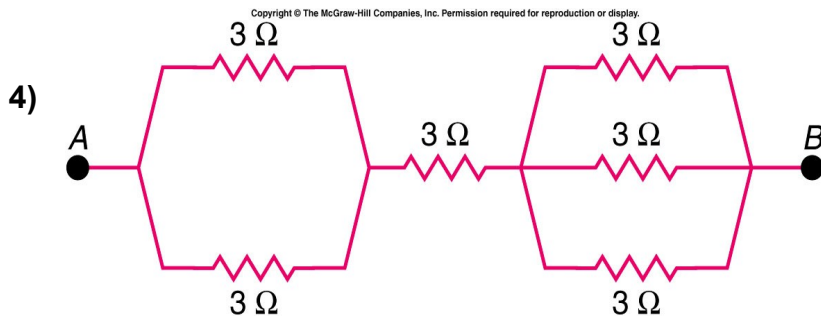
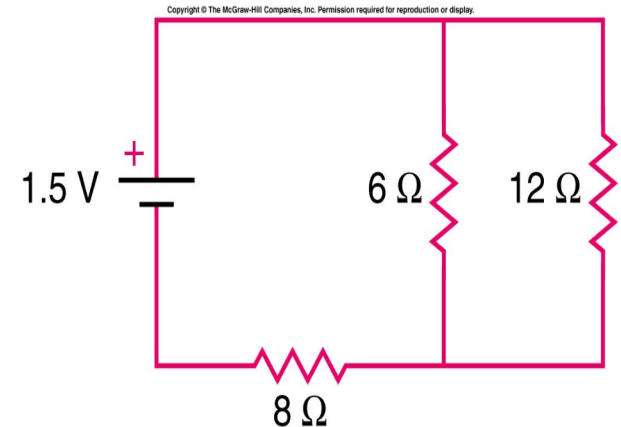
1) resistors are 8, 24, 30, 20 ohms.

In Figure 10-11, the same four resistors as in Figure 10-10 are connected in parallel. Find (a) the combined resistance; (b) the current in the battery; (c) the current in the 8-ohm resistor; and (d) the power dissipated in the 24-ohm resistor.

2) find the main current. And current in each branch.,



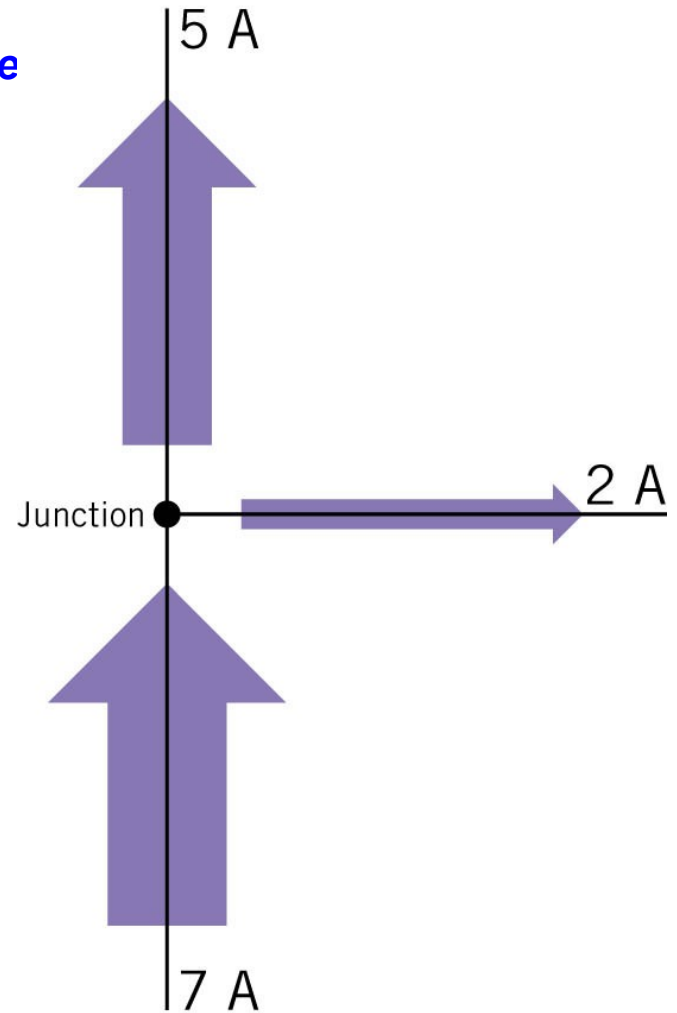
3) find the main current. And current in each branch,



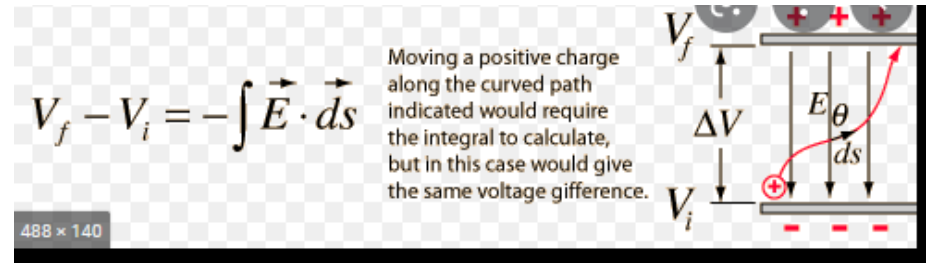
Find the combined resistance

20.10 Kirchhoff's Rule

The junction rule states that the total current directed into a junction must equal the total current directed out of the junction.



The electric force (pulling on charge → electricity) is a conservative force.
 The closed loop integral of $E \cdot ds = 0$ or the work done over a closed loop = 0.



Work by Conservative Forces

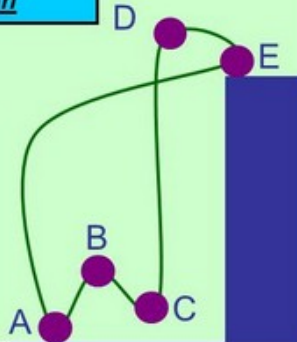
- The work a conservative force does on an object in moving it from a point A to a point B is path independent. It depends only on the path's end points.

- Conservative Forces:

- Gravity
- Elastic (spring) Force
- Electromagnetic Force

For conservative forces it is possible to define a potential energy function

The work done by gravity is the same for path A-B-C-D-E as it is for path A-E



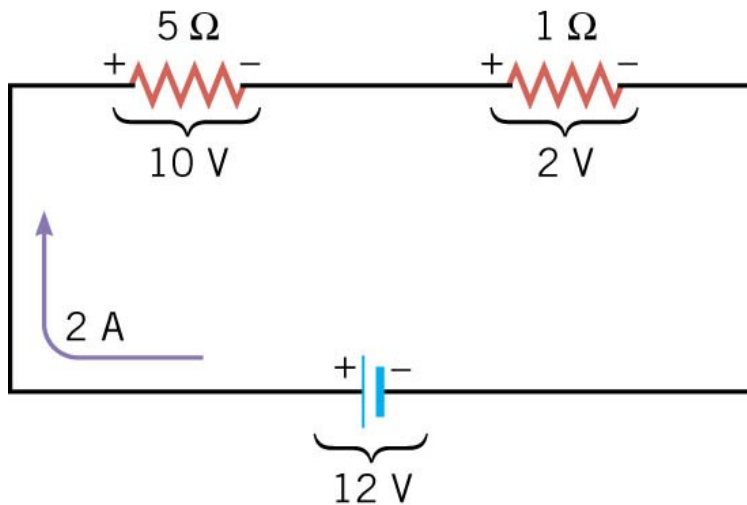
Equivalently, the work done on an object traveling in a closed loop is zero.

$$\mathbf{E} = -\nabla V.$$

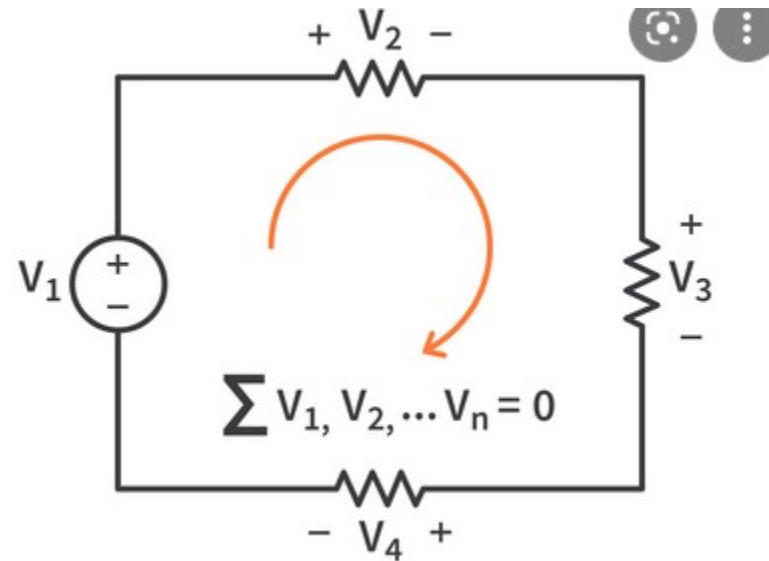
$$\oint \mathbf{E} \cdot d\mathbf{l} = 0,$$

20.10 Kirchhoff's Rules

The loop rule expresses conservation of energy in terms of the electric potential and states that for a closed circuit loop, the total of all potential rises is the same as the total of all potential drops.



$$+12 - 10 - 2 = 0$$



20.10 *Kirchhoff's Rules*

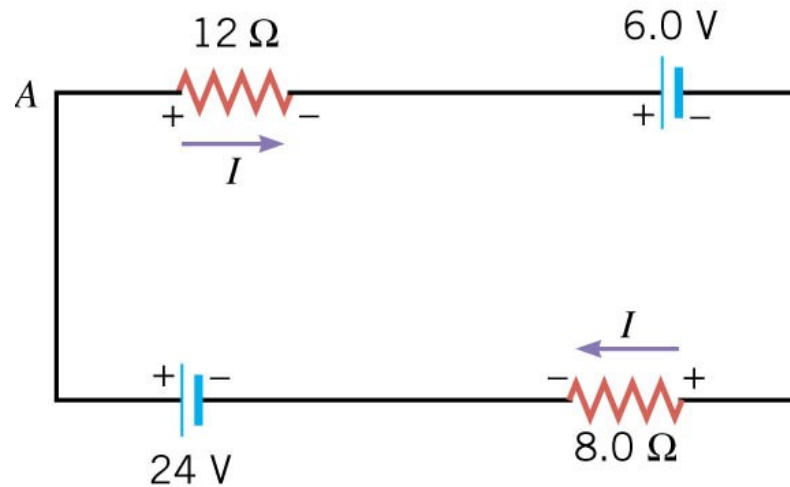
KIRCHHOFF'S RULES

Junction rule. The sum of the magnitudes of the currents directed into a junction equals the sum of the magnitudes of the currents directed out of a junction.

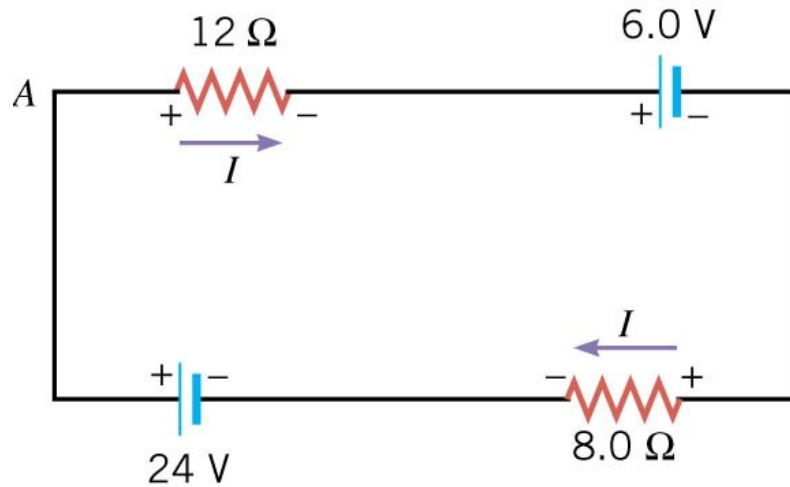
Loop rule. Around any closed circuit loop, the sum of the potential drops equals the sum of the potential rises.

20.10 Kirchhoff's Rules
Example 14 Using Kirchhoff's Loop Rule

Determine the current in the circuit.

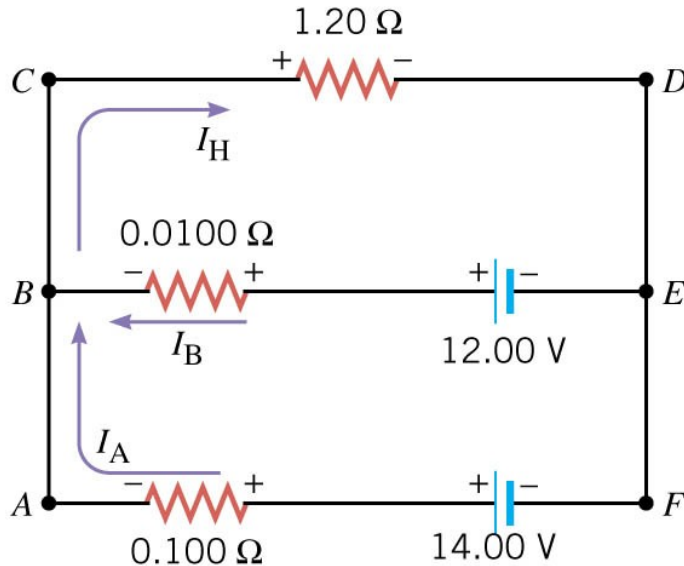
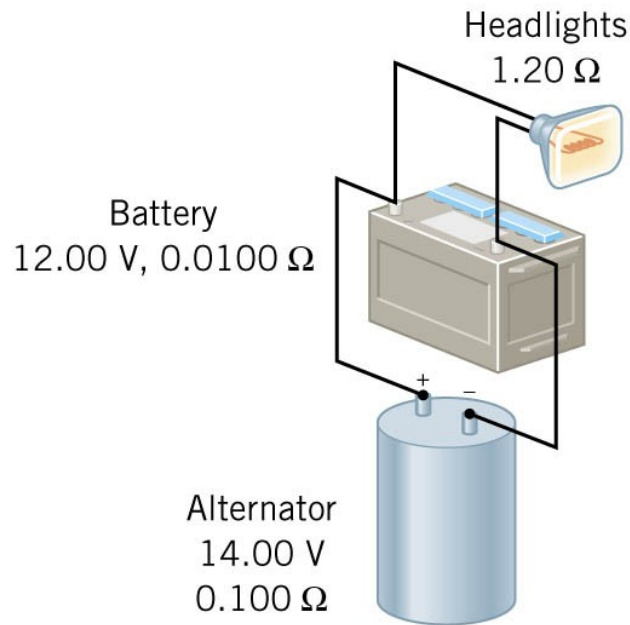


$$-12I - 6 - 8I + 24 = 0$$



$$\underbrace{I(12\ \Omega) + 6.0\ \text{V} + I(8.0\ \Omega)}_{\text{potential drops}} = \underbrace{24\ \text{V}}_{\text{potential rises}}$$

$$I = 0.90\ \text{A}$$



See **Textbook 26.6** and how to use your TI to solve a system of equations
In the matrix form you get:

1	1	-1	I_A	0
0	-0.01	-1.2	I_B	-12
-0.1	0.01	0	I_H	-2

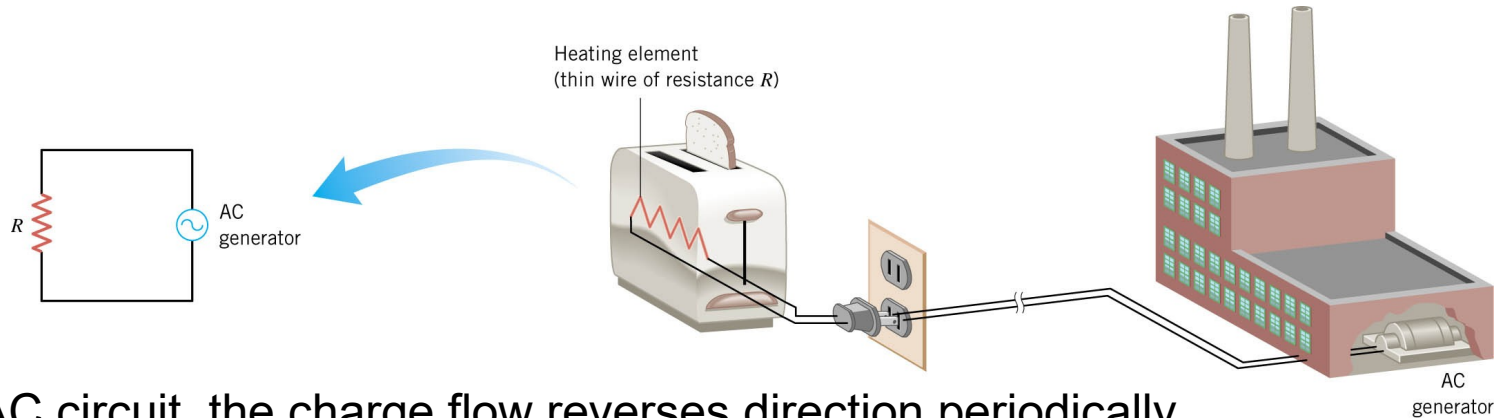
How to use your TI <https://www.youtube.com/watch?v=uGdDFb-hoNc>

20.10 *Kirchhoff's Rules*

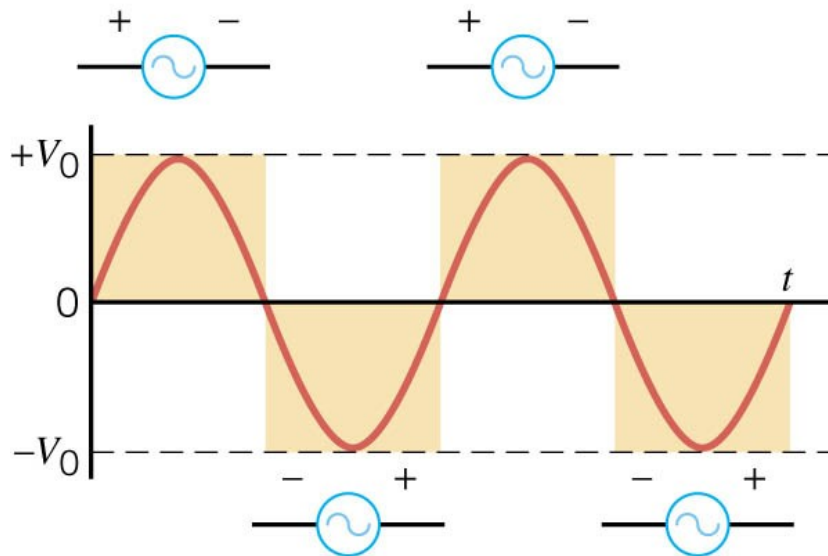
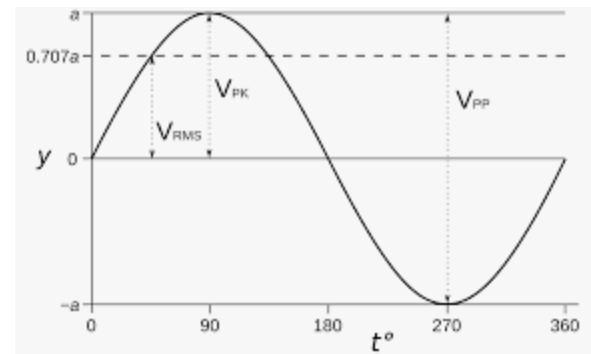
Reasoning Strategy

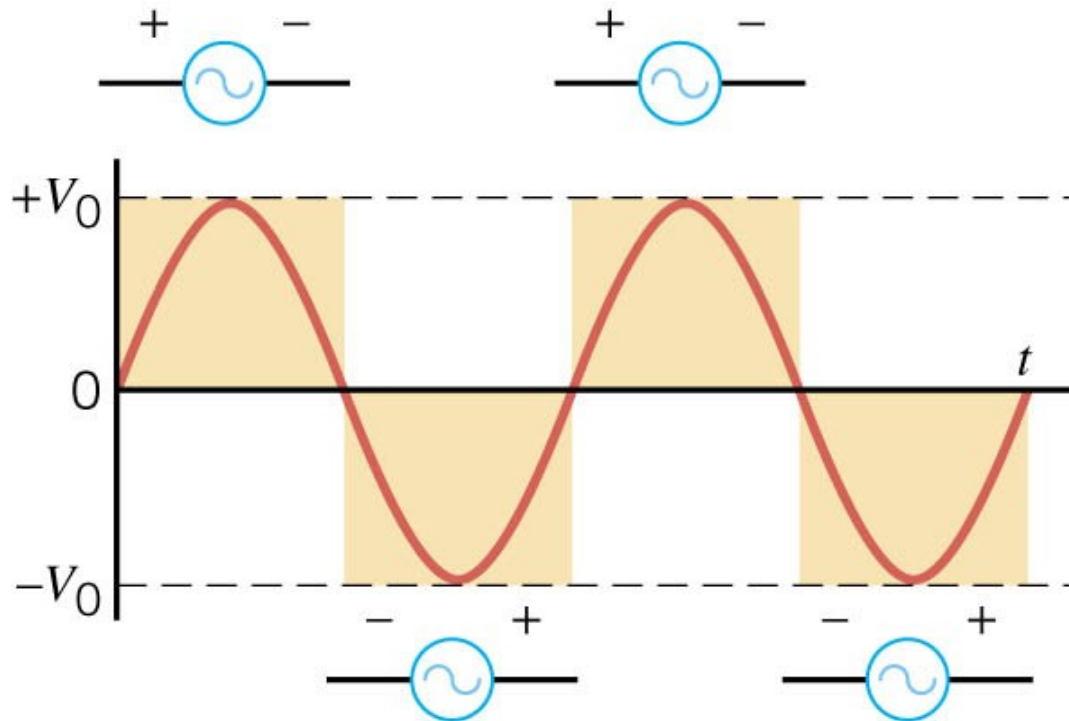
Applying Kirchhoff's Rules

1. Draw the current in each branch of the circuit. Choose any direction. If your choice is incorrect, the value obtained for the current will turn out to be a negative number.
2. Mark each resistor with a + at one end and a – at the other end in a way that is consistent with your choice for current direction in step 1. Outside a battery, conventional current is always directed from a higher potential (the end marked +) to a lower potential (the end marked -).
3. Apply the junction rule and the loop rule to the circuit, obtaining in the process as many independent equations as there are unknown variables.
4. Solve these equations simultaneously for the unknown variables.



In an AC circuit, the charge flow reverses direction periodically.
 In the US the max voltage is about 170V.
 The V_{rms} is 120V (measured by voltmeter).



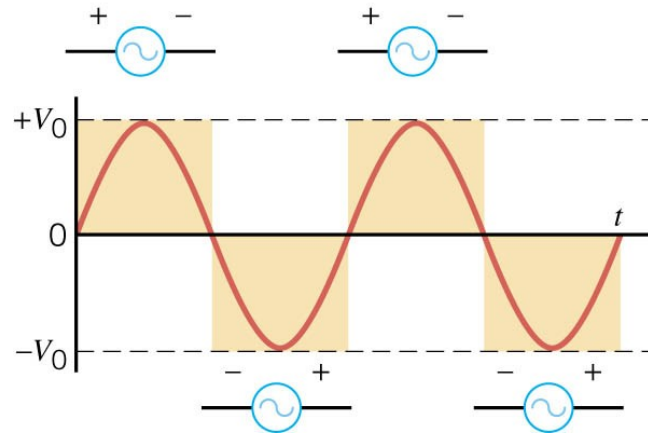


$$V = V_o \sin(2\pi ft)$$

In the US, the frequency is 60Hz
 (1 cycle is $1/60 = 0.017\text{s}$ = called the period).

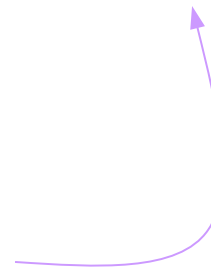
Use wolfram alpha; plot $170 \cdot \sin(2 \cdot \pi \cdot x \cdot 60)$ for x between 0 and 0.05

In circuits that contain only resistance, the current reverses direction each time the polarity of the generator reverses.



$$I = \frac{V}{R} = \frac{V_o}{R} \sin(2\pi ft) = I_o \sin(2\pi ft)$$

peak current

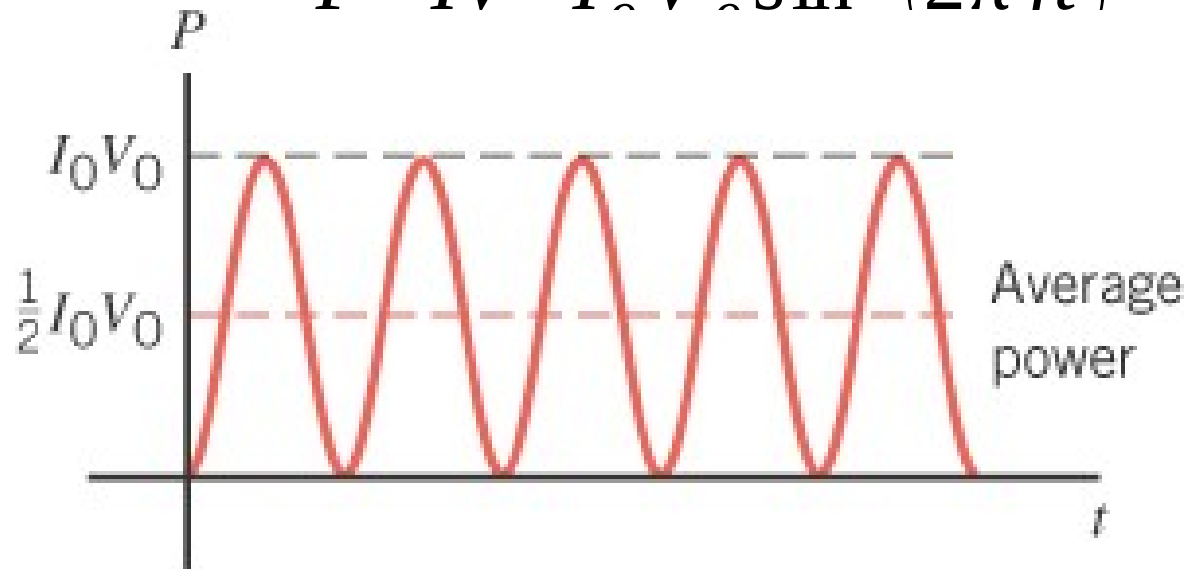


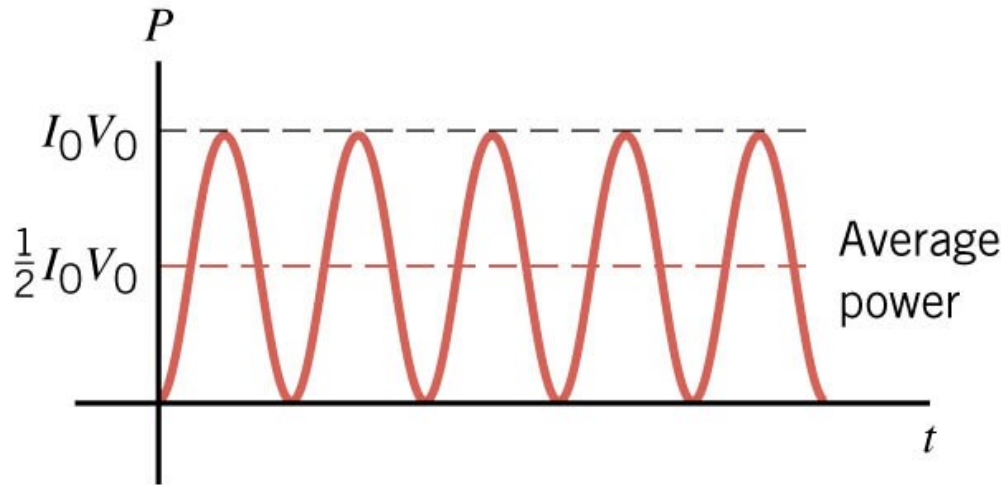
20.5 Alternating Current

$$I = I_o \sin(2\pi ft)$$

$$V = V_o \sin(2\pi ft)$$

$$P = IV = I_o V_o \sin^2(2\pi ft)$$





$$\bar{P} = \frac{I_o V_o}{2} = \left(\frac{I_o}{\sqrt{2}} \right) \left(\frac{V_o}{\sqrt{2}} \right) = I_{rms} V_{rms}$$

The average power is $I_{max} V_{max} / 2$ (average of $\sin^2$ is $\frac{1}{2}$)

$P_{average} = (I_{max} \times \sqrt{2}/2) (V_{max} \times \sqrt{2}/2)$

and $I_{rms} = I_{max} \times \sqrt{2}/2$ $V_{rms} = V_{max} \times \sqrt{2}/2$

When using Ohm's law for AC circuits → use the RMS values.

20.5 *Alternating Current*

$$V_{\text{rms}} = I_{\text{rms}} R$$

$$\bar{P} = V_{\text{rms}} I_{\text{rms}}$$

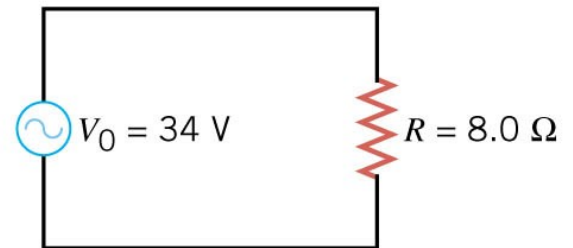
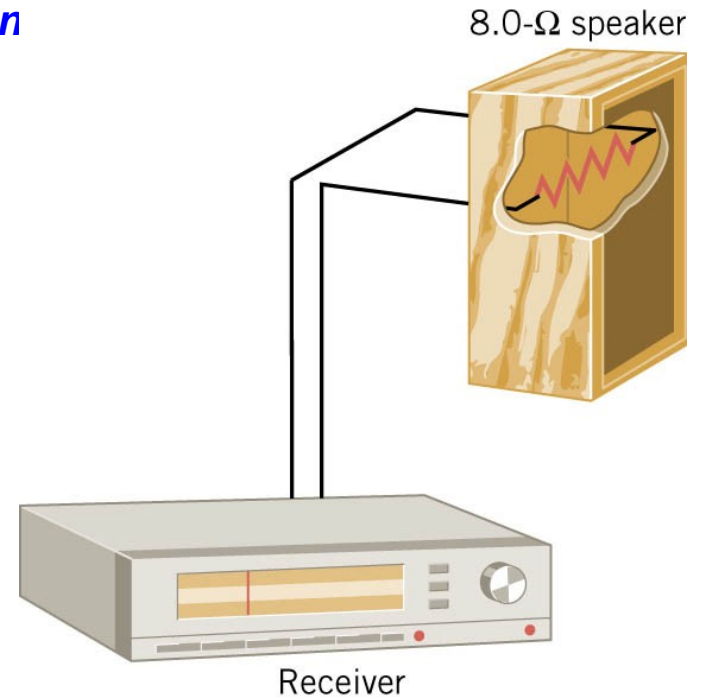
$$\bar{P} = I_{\text{rms}}^2 R$$

$$\bar{P} = \frac{V_{\text{rms}}^2}{R}$$

20.5 Alternating Current

Example 6 Electrical Power Sent to a Loudspeaker

A stereo receiver applies a peak voltage of 34 V to a speaker. The speaker behaves approximately as if it had a resistance of $8.0\ \Omega$. Determine (a) the rms voltage, (b) the rms current, and (c) the average power for this circuit.

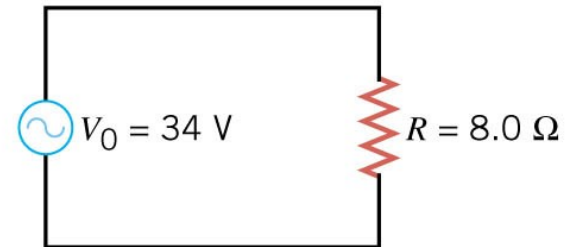
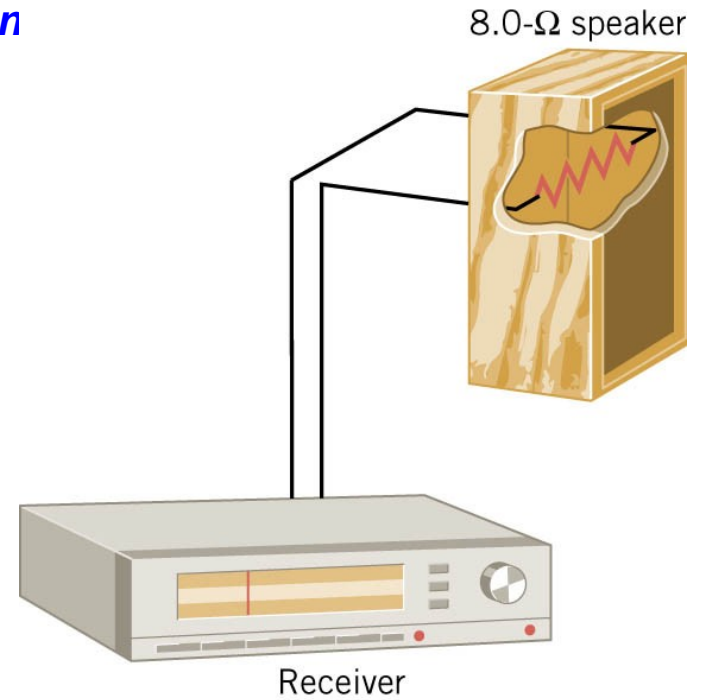


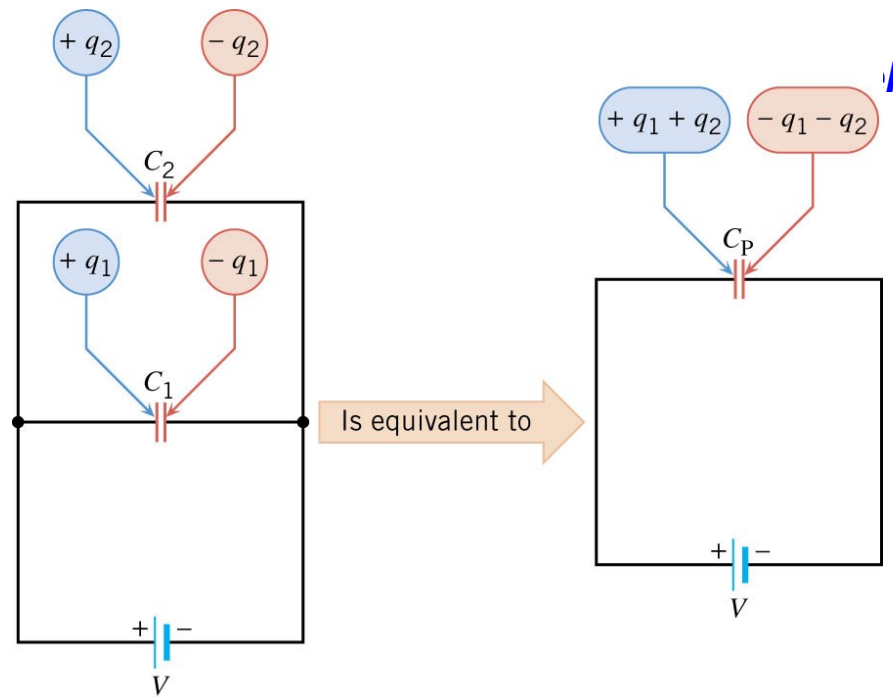
20.5 Alternating Current

(a)
$$V_{\text{rms}} = \frac{V_o}{\sqrt{2}} = \frac{34 \text{ V}}{\sqrt{2}} = 24 \text{ V}$$

(b)
$$I_{\text{rms}} = \frac{V_{\text{rms}}}{R} = \frac{24 \text{ V}}{8.0 \Omega} = 3.0 \text{ A}$$

(c)
$$\bar{P} = I_{\text{rms}} V_{\text{rms}} = (3.0 \text{ A})(24 \text{ V}) = 72 \text{ W}$$

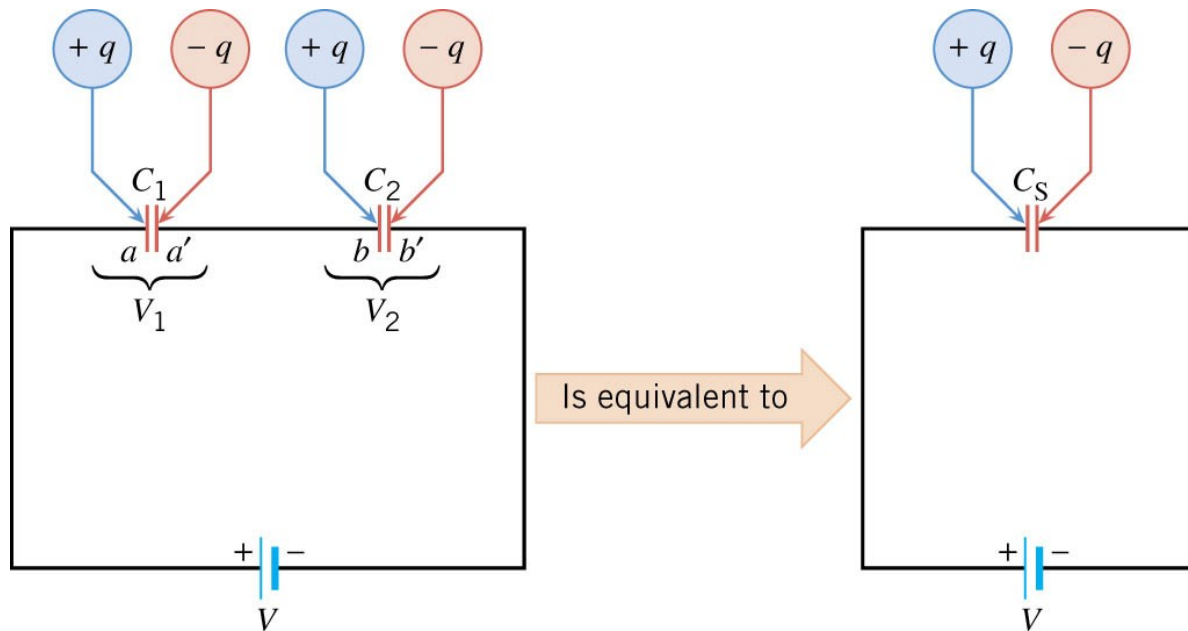




$$q = q_1 + q_2 = C_1 V + C_2 V = (C_1 + C_2) V$$

Parallel capacitors

$$C_P = C_1 + C_2 + C_3 + \dots$$



$$V = V_1 + V_2 = \frac{q}{C_1} + \frac{q}{C_2} = q \left(\frac{1}{C_1} + \frac{1}{C_2} \right)$$

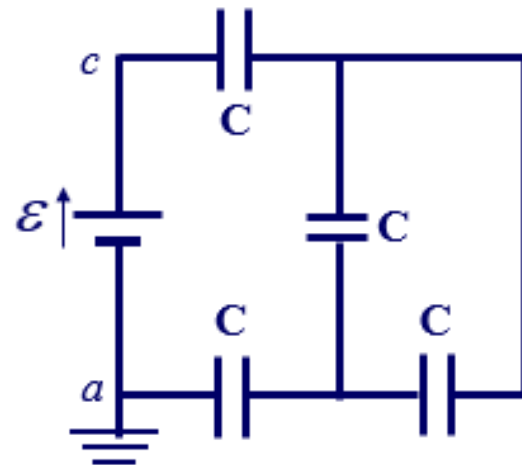
Series capacitors

$$\frac{1}{C_S} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} + \dots$$

Capacitors in series and parallel

3. Four capacitors are connected as shown in figure. Find the equivalent capacitance between points a and c .

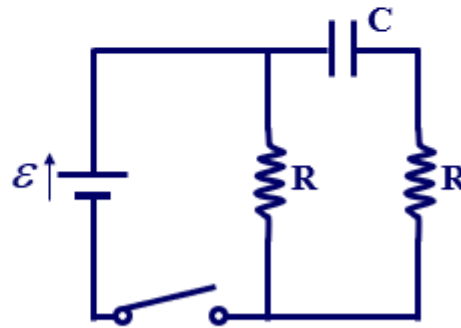
- A. 4 C.
- B. 3 C.
- C. 2.5 C.
- D. 0.4 C.
- E. Cannot determine from information given.



Resistors in series and parallel

7. Consider the circuit shown and assume the battery has no internal resistance. After a very long time, what is the current in the battery?

- A. 0.
- B. $\varepsilon/2R$.
- C. $2\varepsilon/R$.
- D. ε/R .
- E. impossible to determine



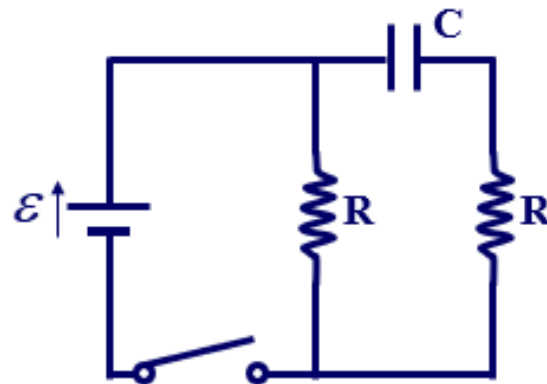
$t=0$, the capacitor can be replaced by a short

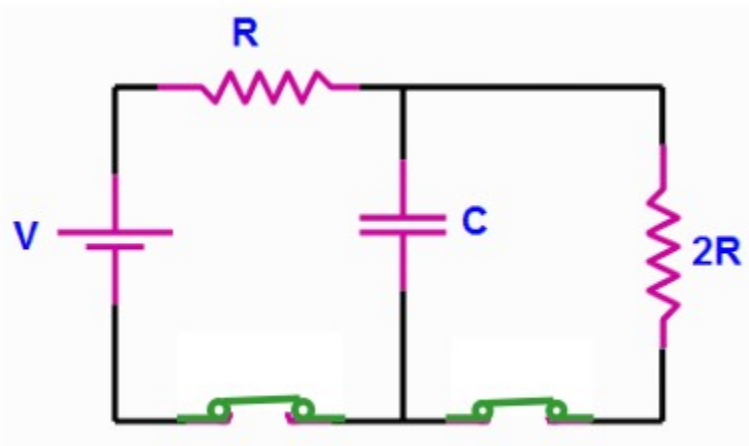
$t=\infty$, the capacitor can be replaced by an open switch

Resistors in series and parallel

6. Consider the circuit shown and assume the battery has no internal resistance. Just after the switch is closed, what is the current in the battery?

- A. 0.
- B. $\varepsilon/2R$.
- C. $2\varepsilon/R$.
- D. ε/R .
- E. impossible to determine

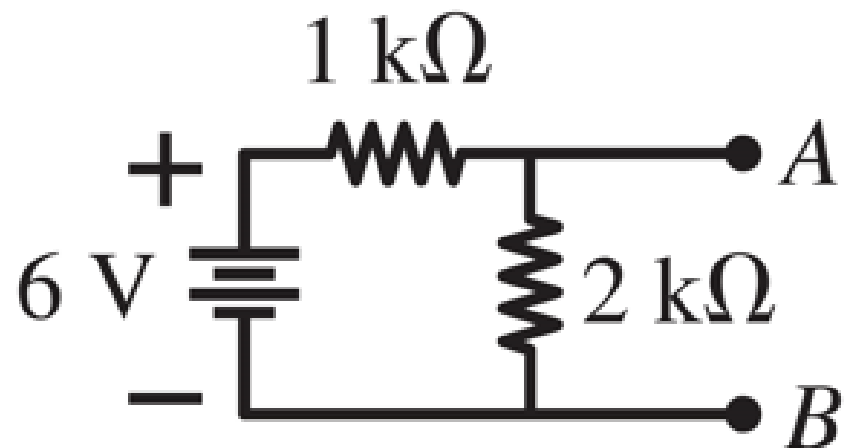




Now suppose both switches are closed. What is the voltage across the capacitor after a very long time?

- A capacitor is charged to 12 V and then connected to points *A* and *B* in the figure, with its positive plate at *A*. What is the current through the 2-k Ω resistor a long time after the capacitor is connected?

- A. 6 mA
- B. 12 mA
- C. 2 mA
- D. 3 mA



Loop Rule The algebraic sum of the changes in potential encountered in a complete traversal of any closed circuit must be zero; see Figs. 5 and 6 and Sample Problem 1.

Junction Rule The sum of the currents entering any junction must be equal to the sum of the currents leaving that junction. See Fig. 8 and the analysis of Section 29-6.

The current in a single-loop circuit containing single resistor R and an emf $\mathcal{E}$ with internal resistance r (see Fig. 5) is

$$i = \frac{\mathcal{E}}{R + r}, \quad [4]$$

which reduces to Eq. 2 when $r = 0$.

Resistors, and other circuit elements, are in *series* if the sum of the individual potential differences is the potential difference of the combination; see Fig. 6. The equivalent resistance of resistors in series is

$$R_{\text{eq}} = \sum_n R_n \quad (\text{resistors in series}). \quad [7]$$

Resistors, and other circuit elements, are in *parallel* if they share a common potential difference; see Fig. 7. The equivalent resistance of resistors in parallel is

$$R_{\text{eq}} = \sum_n \frac{1}{R_n} \quad (\text{resistors in parallel}). \quad [20]$$

Sample Problem 3 illustrates the use of the circuit rules and Sample Problem 4 shows an interesting application of the analysis of a circuit using equivalent resistances of series and parallel combinations. Section 29-7 uses the circuit rules to analyze the operation of conventional ammeters, voltmeters, and potentiometers.

When an emf $\mathcal{E}$ is applied to a resistor R and capacitor C in series, as in Fig. 15, the charge on the capacitor increases according to

$$q = C\mathcal{E}(1 - e^{-t/RC}) \quad (\text{charging capacitor}), \quad [29]$$

in which $C\mathcal{E} = q_0$ is the equilibrium charge and $RC = \tau$ is the *capacitive time constant* of the circuit. When the emf is replaced by a short circuit, the charge on the capacitor decays according to

$$q = q_0 e^{-t/RC} \quad (\text{discharging capacitor}). \quad [34]$$